
TRIGONOMETRY

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TABLE OF CONTENTS

TABLE OF CONTENTS	2	2. HEIGHTS AND DISTANCES.....	31
1. RIGHT TRIANGLE TRIGONOMETRY	3	2.1 Heights and Distances: Basics	31
1.1 Trigonometric Ratios	3	2.2 Heights and Distances: II	42
1.2 Trigonometric Ratios-II	10	2.3 3D Problem Solving	50
1.3 Triangles	12	2.4 Bearings and Coordinate Geometry	59
1.4 Polygons	21		

1. RIGHT TRIANGLE TRIGONOMETRY

1.1 Trigonometric Ratios

A. The Greek Letters

You might have already see the Greek Letter π used in the formulas related to circles:

$$\begin{aligned} \text{Area} &= A = \pi r^2 \\ C &= \text{Circumference} = 2\pi r \end{aligned}$$

1.1: Greek Letters

In trigonometry, we often use Greek Letters for measures of angles. Some of the common ones are mentioned below:

$$\begin{aligned} \alpha &= \text{alpha} \\ \beta &= \text{beta} \\ \gamma &= \text{gamma} \\ \phi &= \text{phi} \\ \sigma &= \text{sigma} \\ \theta &= \text{theta} \end{aligned}$$

B. The Trigonometric Functions

Trigonometry was first used by the Egyptians. They lived next to the river Nile, and every year when the Nile flooded their agricultural marking were lost, which distinguished one field from another. To recover the marking, they developed a sophisticated system of geometry including trigonometry.

Trigonometry is based on the study of triangles. Initially, right-angled triangles, but then building up and generalising. Apart from geometry, trigonometry has applications in many fields.

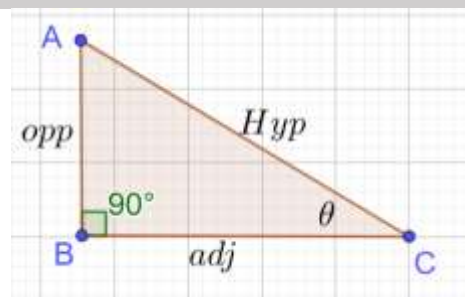
1.2: Defining the Trigonometric Functions

If we have a right-angled triangle with one angle with a measure of θ (see diagram), then we can define certain ratios that are very useful.

$$\sin \theta = \frac{\text{opp}}{\text{hyp}}, \quad \cos \theta = \frac{\text{adj}}{\text{hyp}}, \quad \tan \theta = \frac{\sin \theta}{\cos \theta} = \frac{\text{opp}}{\text{adj}}$$

We can also define the reciprocals of these trigonometric functions:

$$\csc \theta = \frac{1}{\sin \theta}, \quad \sec \theta = \frac{1}{\cos \theta}, \quad \cot \theta = \frac{1}{\tan \theta}$$



1.3: The Six Trigonometric Functions

The six trigonometric functions are:

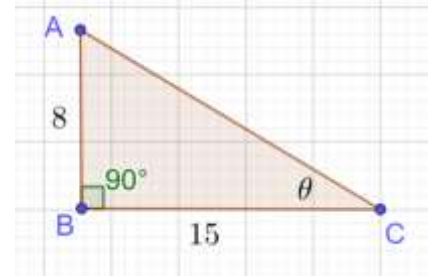
$$\begin{aligned} \sin \theta, \cos \theta, \tan \theta \\ \csc \theta, \sec \theta, \cot \theta \end{aligned}$$

1.4: Mnemonic: Reciprocal Functions

Obtaining the reciprocal of $\sin \theta$ and $\cos \theta$ changes the initial letter.

$$\csc \theta = \frac{1}{\sin \theta} = \frac{\text{hyp}}{\text{opp}}$$

$$\sec \theta = \frac{1}{\cos \theta} = \frac{\text{hyp}}{\text{adj}}$$



Example 1.5

Find the six trigonometric ratios of θ for the diagram shown alongside.

Part A

In $\triangle ABC$, by the Pythagorean Theorem:

$$AB = 8, BC = 15 \Rightarrow AC = 17 \text{ (Pythagorean Triplet)}$$

Now, we can use the definitions of the trigonometric functions to calculate their values:

$$\sin \theta = \frac{\text{opp}}{\text{hyp}} = \frac{8}{17} \Rightarrow \csc \theta = \frac{1}{\sin \theta} = \frac{17}{8}$$

$$\cos \theta = \frac{\text{adj}}{\text{hyp}} = \frac{15}{17} \Rightarrow \sec \theta = \frac{1}{\cos \theta} = \frac{17}{15}$$

$$\tan \theta = \frac{\text{opp}}{\text{adj}} = \frac{8}{15} \Rightarrow \cot \theta = \frac{1}{\tan \theta} = \frac{15}{8}$$

Example 1.6

Find the value(s) of the six trigonometric functions for the smaller angle of a right-angled triangle with two sides 5 and 12.

Case I

Suppose the sides which are 5 and 12 are the legs of the right-angled triangle.

Then, by Pythagoras Theorem:

$$\text{Hyp} = \sqrt{5^2 + 12^2} = \sqrt{25 + 144} = \sqrt{169} = 13$$

Let $\angle ACB = \theta$:

$$\sin \theta = \frac{\text{opp}}{\text{hyp}} = \frac{5}{13}$$

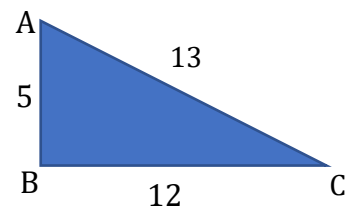
$$\cos \theta = \frac{\text{adj}}{\text{hyp}} = \frac{12}{13}$$

$$\tan \theta = \frac{\text{opp}}{\text{adj}} = \frac{5}{12}$$

$$\csc \theta = \frac{\text{hyp}}{\text{opp}} = \frac{13}{5}$$

$$\sec \theta = \frac{\text{hyp}}{\text{adj}} = \frac{13}{12}$$

$$\cot \theta = \frac{\text{adj}}{\text{opp}} = \frac{12}{5}$$



Case II

12 is the hypotenuse $\Rightarrow$ Legs are 5 and $\sqrt{119}$

1.7: Primitive Pythagorean Triplets

- Three integers which can be the sides of a right triangle form a Pythagorean Triplet.
- A Pythagorean Triplet (a, b, c) where $HCF(a, b, c) = 1$ is called a primitive Pythagorean Triplet.

Primitive Pythagorean Triplets			
3, 4, 5	20, 21, 29	11, 60, 61	36, 77, 85
5, 12, 13	12, 35, 37	33, 56, 65	13, 84, 85
8, 15, 17	9, 40, 41	16, 63, 65	39, 80, 89
7, 24, 25	28, 45, 53	48, 55, 73	65, 72, 97

$$2(3,4,5) = (6,8,10)$$

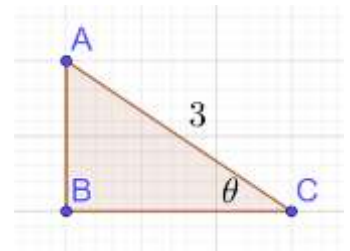
1.8: Reverse Calculations

Given the value of θ , we can use the trigonometric functions to find the length of side of a triangle.

Example 1.9

If the angle shown in the diagram has a measure of θ° , then find, in terms of θ , the length of

- Side AB
- Side BC



$$\sin \theta = \frac{\text{opp}}{\text{hyp}} = \frac{AB}{3} \Rightarrow AB = 3 \sin \theta$$

$$\cos \theta = \frac{BC}{3} \Rightarrow BC = 3 \cos \theta$$

1.10: Reference Triangle

If we are given one of the trigonometric ratios, we can draw a triangle with two sides that meet these conditions, use the Pythagorean theorem to find the third side, and then find the remaining trigonometric ratios.

- A reference triangle is a triangle that meets the conditions for the information that you have.

Example 1.11

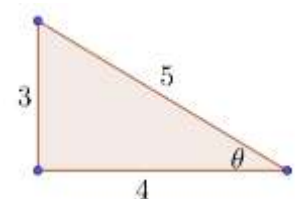
In each case, find the other five trigonometric ratios.

- $\sin \theta = \frac{3}{5}$
- $\sin \theta = \frac{24}{25}$

Part A

$$\sin \theta = \frac{3}{5} = \frac{\text{opp}}{\text{hyp}}$$

Draw a *reference triangle* (see diagram) with
 $\text{opp} = 3, \text{hyp} = 5$



Now, we have a Pythagorean Triplet:

$$(3,4,5)$$

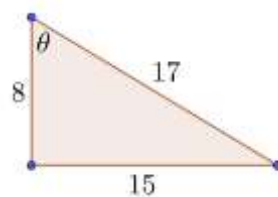
Now that we have all three sides, the six trigonometric ratios are:

$$\cos \theta = \frac{4}{5}, \tan \theta = \frac{3}{4}, \csc \theta = \frac{1}{\sin \theta} = \frac{5}{3}, \sec \theta = \frac{1}{\cos \theta} = \frac{5}{4}, \cot \theta = \frac{4}{3}$$

Part C

Draw a right triangle, and then $\text{opp} = 24$, $\text{hyp} = 25 \Rightarrow \text{adj} = 7$

$$\cos \theta = \frac{7}{25}, \tan \theta = \frac{24}{7}, \sec \theta = \frac{1}{\sin \theta} = \frac{25}{24}, \csc \theta = \frac{1}{\cos \theta} = \frac{25}{7}, \cot \theta = \frac{7}{24}$$



Example 1.12

In each case, find the other five trigonometric ratios.

$$\sin \theta = \frac{p}{q}$$

1.13: Trig Functions for 45°

$$\begin{aligned}\sin 45^\circ &= \frac{1}{\sqrt{2}} = \frac{\sqrt{2}}{2} \approx 0.70711 \\ \cos 45^\circ &= \frac{1}{\sqrt{2}} \\ \tan 45^\circ &= 1\end{aligned}$$

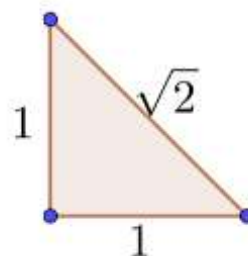
Draw an isosceles right triangle with angles 45° , 45° , and 90° . This kind of triangle is called:

$45 - 45 - 90$ Triangle

Suppose:

$$AB = 1 \Rightarrow BC = 1 \Rightarrow AC = \sqrt{1^2 + 1^2} = \sqrt{2}$$

Now, the values can be obtained from the triangle by using the definitions.



1.14: Trig Functions for 30° and 60°

$$\begin{aligned}\sin 60^\circ &= \frac{\sqrt{3}}{2}, & \cos 60^\circ &= \frac{1}{2}, & \tan 60^\circ &= \sqrt{3} \\ \sin 30^\circ &= \frac{1}{2}, & \cos 30^\circ &= \frac{\sqrt{3}}{2}, & \tan 30^\circ &= \frac{1}{\sqrt{3}}\end{aligned}$$

Draw equilateral $\triangle ACB$ with side 1.

Drop a perpendicular from vertex A, meeting the opposite side at point X. Then:

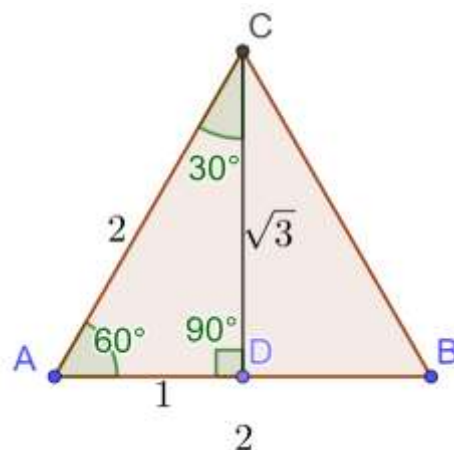
$\triangle ACD$ is a $30 - 60 - 90$ triangle.

In an isosceles triangle, the altitude is also the median:

$$AD = DB = \frac{1}{2}$$

In right $\triangle ABX$, by the Pythagorean Theorem:

$$AX = h = \sqrt{AB^2 - BX^2} = \sqrt{1^2 - \left(\frac{1}{2}\right)^2} = \sqrt{\frac{3}{4}} = \frac{\sqrt{3}}{2}$$



Now, the values can be obtained from the triangle by using the definitions.

Example 1.15

Find the exact value:

- A. $\cos 30^\circ$
- B. $\cot 30^\circ$
- C. $\csc 60^\circ$
- D. $\cos 60^\circ$
- E. $\tan 45^\circ$
- F. $\sec 60^\circ$
- G. $\cos 45^\circ$

$$\begin{aligned}\cos 30^\circ &= \frac{\sqrt{3}}{2} \\ \cot 30^\circ &= \frac{1}{\tan 30^\circ} = \frac{1}{\frac{1}{\sqrt{3}}} = \sqrt{3} \\ \csc 60^\circ &= \frac{1}{\sin 60^\circ} = \frac{1}{\frac{\sqrt{3}}{2}} = \frac{2}{\sqrt{3}}\end{aligned}$$

$$\begin{aligned}\cos 60^\circ &= \frac{1}{2} \\ \tan 45^\circ &= 1 \\ \sec 60^\circ &= \frac{1}{\cos 60^\circ} = \frac{1}{\frac{1}{2}} = 2 \\ \cos 45^\circ &= \frac{1}{\sqrt{2}}\end{aligned}$$

C. Inverse Trigonometry Functions

Example 1.16

Find the value of θ if $\cos \theta = \frac{\sqrt{3}}{2}$.

$$\cos \theta = \frac{\sqrt{3}}{2} \Rightarrow \theta = 30^\circ$$

1.17: Inverse Trigonometry Functions

The trigonometric functions associate a value with an angle. The inverse trigonometric functions do the reverse process.

Example 1.18

Find the value of θ if $\cos \theta = \frac{\sqrt{3}}{2}$.

$$\cos \theta = \frac{\sqrt{3}}{2}$$

Apply the $\cos^{-1}$ function on both sides:

$$\theta = \cos^{-1}\left(\frac{\sqrt{3}}{2}\right) = 30^\circ$$

Example 1.19

Find θ in each case below:

- A. $\cot \theta = \sqrt{3}$

- B. $\csc \theta = \frac{2}{\sqrt{3}}$
- C. $\cos \theta = \frac{1}{2}$
- D. $\tan \theta = 1$

$$\begin{aligned}\cot \theta &= \sqrt{3} \Rightarrow \theta = \cot^{-1} \sqrt{3} = 30^\circ \\ \csc \theta &= \frac{2}{\sqrt{3}} \Rightarrow \sin \theta = \frac{\sqrt{3}}{2} \Rightarrow \theta = \sin^{-1} \left(\frac{\sqrt{3}}{2} \right) = 60^\circ \\ \cos \theta &= \frac{1}{2} \Rightarrow \theta = \cos^{-1} \left(\frac{1}{2} \right) = 60^\circ \\ \tan \theta &= 1 \Rightarrow \theta = \tan^{-1} 1 = 45^\circ\end{aligned}$$

Example 1.20

Solve for θ :

- A. $\sin \theta = x$
- B. $\cos \theta = y$
- C. $\tan \theta = z$

$$\begin{aligned}\theta &= \sin^{-1} x \\ \theta &= \cos^{-1} y \\ \theta &= \tan^{-1} z\end{aligned}$$

D. Co-Function Identities

1.21: Co-function Identities

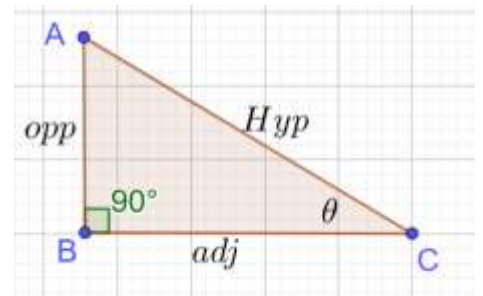
$$\begin{aligned}\sin(90 - \theta) &= \cos \theta \\ \cos(90 - \theta) &= \sin \theta\end{aligned}$$

In Right $\triangle ABC$, right – angled at B

$$\text{Let } \angle C = \theta \Rightarrow \angle A = (90 - \theta)$$

$$\sin \angle A = \sin \angle(90 - \theta) = \frac{\text{opp}}{\text{hyp}} = \frac{BC}{AC} = \cos \theta$$

$$\cos \angle A = \cos \angle(90 - \theta) = \frac{\text{adj}}{\text{hyp}} = \frac{AB}{AC} = \sin \theta$$



Example 1.22

- A. $\cos 43^\circ - \sin 47^\circ$

Rewrite $43 = 90 - 47$:

Use $\cos(90 - \theta) = \sin \theta$:

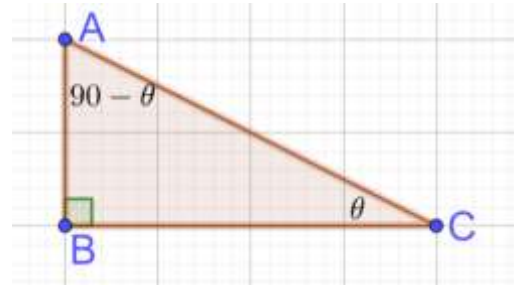
$$\begin{aligned}\cos 43^\circ - \sin 47^\circ \\ &= \cos(90^\circ - 47^\circ) - \sin 47^\circ \\ &= \sin 47^\circ - \sin 47^\circ = 0\end{aligned}$$

1.23: $\cos(90 - \theta) = \sin \theta$

Right Triangle Trigonometry Proof

$$\sin \theta = \cos(90 - \theta) = \frac{AB}{AC}$$

$$\cos \theta = \sin(90 - \theta) = \frac{BC}{AC}$$



Second Quadrant Proof

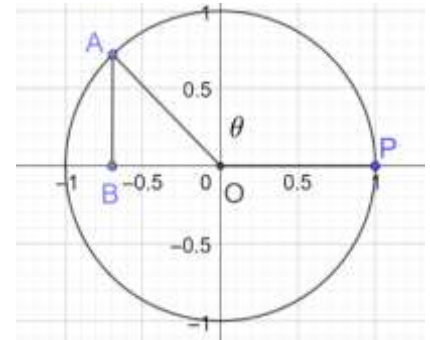
Coordinates of A

$$= A(x, y) = A(\cos \theta, \sin \theta)$$

$$\angle AOP = \theta$$

$$\angle AOB = 180 - \theta$$

$$\angle BAO = 90 - (180 - \theta) = \theta - 90$$



Using $\cos \theta = \cos(-\theta)$:

$$\cos(90 - \theta) = \cos(\theta - 90)$$

Equation I

From $\triangle AOB$:

$$\cos(\theta - 90) = \frac{\text{adj}}{\text{hyp}} = \frac{AB}{1} = AB = \sin \theta$$

Equation II

$$\cos(90 - \theta) = \sin \theta$$

1.24: $\tan(90 - \theta) = \cot \theta$

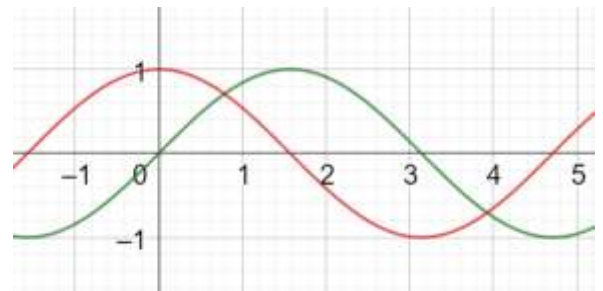
$$\tan(90 - \theta) = \frac{\sin(90 - \theta)}{\cos(90 - \theta)} = \frac{\cos \theta}{\sin \theta} = \cot \theta$$

1.25: Translation

$$\sin(90 + \theta) = \cos \theta$$

The graph of $y = \sin x$ when translated to the left 90° gives

$$y = \cos x$$



E. Degree-Minute-Second Notation

Example 1.26: Degree-Minute-Second Notation

The degree-minute second notation is a notation for precise communication of values of angles. It is not so common now.

$$\underbrace{1^\circ}_{\text{Degrees}} = \underbrace{60'}_{\text{Minutes}} = \underbrace{3600''}_{\text{Seconds}}$$

1 Degree = 60 Minutes

1 Minute = 60 Seconds

Convert from Degree-Minute-Second notation to Decimal Notation

A. $6^{\circ}15'30''$

Convert from Decimal Notation to Degree-Minute-Second notation

B.

$$6^{\circ}15'30'' = 6^{\circ}15' + \frac{30'}{60} = 6^{\circ}15.5' = 6^{\circ} + \frac{15.5'}{60} = 6^{\circ} + \left(\frac{15}{60}\right)^{\circ} + \left(\frac{0.5}{60}\right)^{\circ} = 6^{\circ} + 0.25^{\circ} + \left(\frac{1}{120}\right)^{\circ}$$

*1.2 Trigonometric Ratios-II

A. Some Further Special Triangles

In this section, we cover trigonometric ratios which are used less frequently.

Sometimes these ratios are required to solve a problem. Sometimes, they are not required, but make the solution shorter if you use them.

1.27: Trig Functions of 22.5° and 67.5°

$$\begin{aligned}\sin 22.5^{\circ} &= \frac{\sqrt{2} - 1}{\sqrt{4 - 2\sqrt{2}}} \\ \cos 22.5^{\circ} &= \frac{1}{\sqrt{4 - 2\sqrt{2}}} \\ \tan 22.5^{\circ} &= \sqrt{2} - 1\end{aligned}$$

Draw Rectangle $ABCD$ with length $\sqrt{2}$ and width 1.

On side BC , pick point P such that $BP = 1$.

In isosceles $\triangle ABP$:

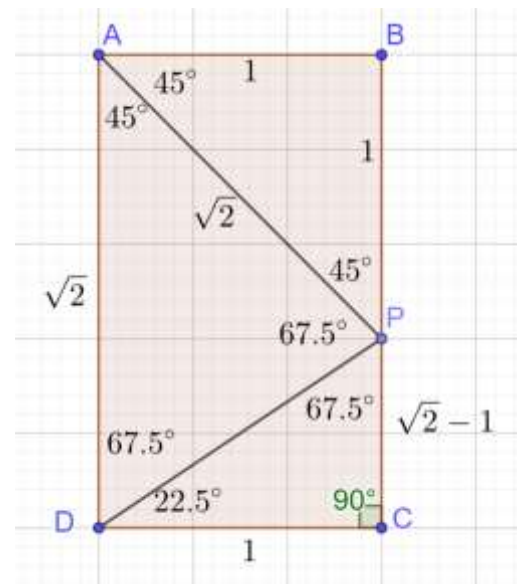
$$AP = \sqrt{2}(AB) = \sqrt{2}$$

In isosceles $\triangle DAP$

$$\angle ADP = \frac{180 - 45}{2} = \frac{135}{2} = 67.5^{\circ}$$

In $\triangle PDC$

$$\begin{aligned}\angle PDC &= 90 - 67.5 = 22.5^{\circ} \\ \angle DPC &= 180 - 90 - 22.5 = 67.5^{\circ} \\ DC &= AB = 1 \\ PC &= BC - BP = \sqrt{2} - 1\end{aligned}$$



By Pythagoras Theorem, in $\triangle PDC$

$$DP = \sqrt{1^2 + (\sqrt{2} - 1)^2} = \sqrt{1 + 2 - 2\sqrt{2} + 1} = \sqrt{4 - 2\sqrt{2}}$$

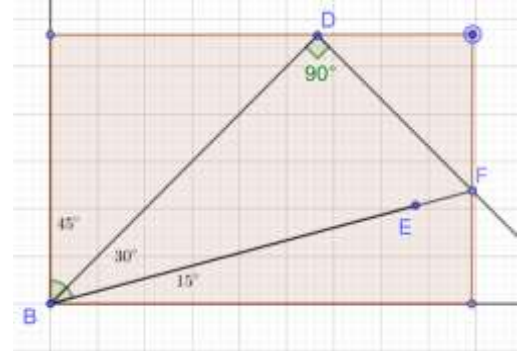
1.28: Trig Functions of 15° and 75°

$$\sin 15^{\circ} = \cos 75^{\circ} = \frac{\sqrt{3} - 1}{2\sqrt{2}}$$

$$\begin{aligned}\cos 15^\circ &= \sin 75^\circ = \frac{\sqrt{3} + 1}{2\sqrt{2}} \\ \tan 15^\circ &= \cot 75^\circ = 2 - \sqrt{3} \\ \tan 75^\circ &= \cot 15^\circ = 2 + \sqrt{3}\end{aligned}$$

Draw

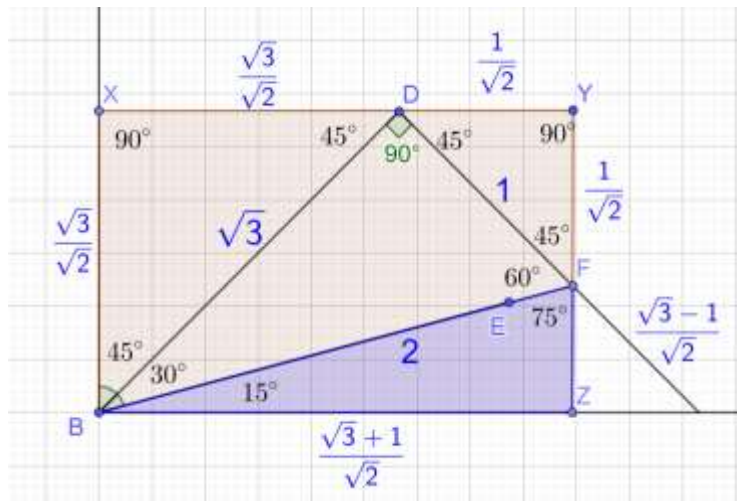
- $\angle ABC$ with 90°
- $\angle ABD$ with 45°
- $\angle DBE$ with 30°
- $\angle BDF$ with 90° intersecting BE at F .
- Rectangle $XYZB$ such that the left lies on the y-axis, the bottom lies on the x-axis, D lies on the top, and F lies on the right side.



$$\angle XDB = 180 - 90 - 45 = 45^\circ$$

In $45 - 45 - 90 \triangle XDB$:

$$BX = XD = \frac{\sqrt{3}}{\sqrt{2}}$$



$$\angle FDY = 180 - 45 - 90 = 45^\circ$$

$$\angle YFD = 180 - 45 - 90 = 45^\circ$$

In $45 - 45 - 90 \triangle DYF$:

$$DY = YF = \frac{1}{\sqrt{2}}$$

$$BZ = XD + DY = \frac{\sqrt{3}}{\sqrt{2}} + \frac{1}{\sqrt{2}} = \frac{\sqrt{3} + 1}{\sqrt{2}}$$

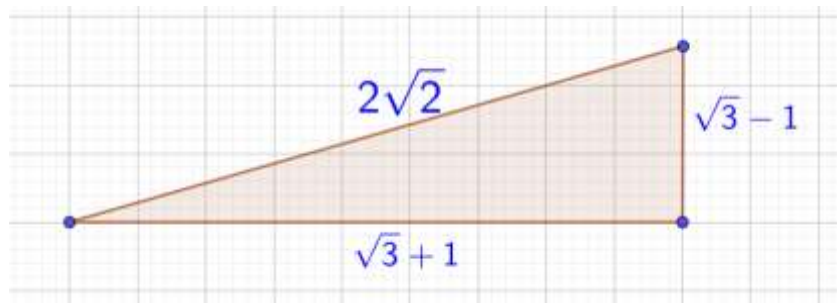
$$FZ = XB - YF = \frac{\sqrt{3}}{\sqrt{2}} - \frac{1}{\sqrt{2}} = \frac{\sqrt{3} - 1}{\sqrt{2}}$$

And note that

$$\angle FBZ = 90 - 45 - 30 = 15^\circ$$

$$\angle BFZ = 180 - 90 - 15 = 75^\circ$$

Hence, $\triangle BFZ$ is $15^\circ - 75^\circ - 90^\circ$



$$\begin{aligned}\sin 15^\circ &= \cos 75^\circ = \frac{\sqrt{3} - 1}{2\sqrt{2}} \\ \cos 15^\circ &= \sin 75^\circ = \frac{\sqrt{3} + 1}{2\sqrt{2}} \\ \tan 15^\circ &= \frac{\sqrt{3} - 1}{\sqrt{3} + 1} = \frac{\sqrt{3} - 1}{\sqrt{3} + 1} \cdot \frac{\sqrt{3} - 1}{\sqrt{3} - 1} = \frac{3 - 2\sqrt{3} + 1}{2} = \frac{4 - 2\sqrt{3}}{2} = 2 - \sqrt{3} \\ \cot 15^\circ &= \frac{1}{2 - \sqrt{3}} = \frac{2 + \sqrt{3}}{1} = 2 + \sqrt{3} \\ \tan 75^\circ &= \frac{1}{\tan 15^\circ} = \frac{1}{2 - \sqrt{3}} = 2 + \sqrt{3}\end{aligned}$$

1.3 Triangles

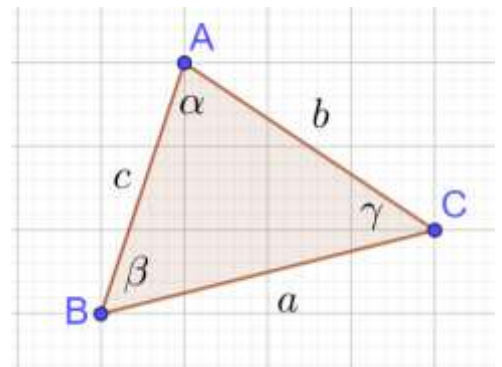
A. Basics

Given some side lengths and angles related to a right triangle, the process of finding the remaining values is called “solving” the right triangle. This can be accomplished using the trigonometrical ratios.

1.29: Triangle Nomenclature

It is *traditional* that for a general triangle diagram:

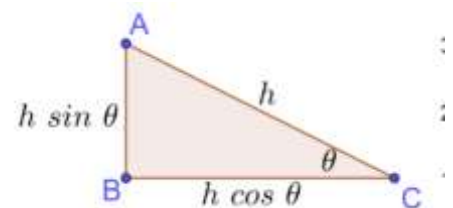
- We use capital letters for the vertices A, B, C
- We use small letters for the lengths of the sides opposite the vertices a, b, c
- We use Greek letters for the angles α, β, γ



Example 1.30

A right-angled triangle has an angle of $\theta, \theta \neq 90^\circ$. Find the lengths of the other two sides in terms of its hypotenuse h .

$$\begin{aligned}\sin \theta &= \frac{AB}{h} \Rightarrow AB = h \sin \theta \\ \cos \theta &= \frac{BC}{h} \Rightarrow BC = h \cos \theta\end{aligned}$$



1.31: Solving a Triangle

Finding the values of the angles and the sides of a triangle is called “solving” the triangle.

Example 1.32

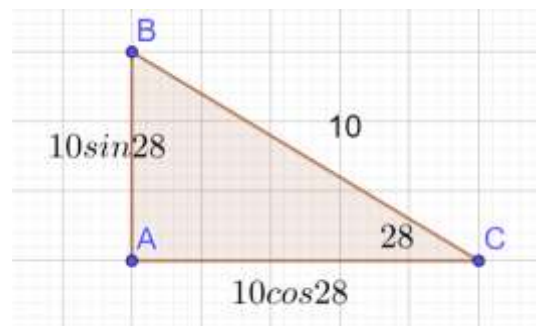
A right-angled triangle ABC , right-angled at A , has $\angle BCA = 28^\circ$. If $BC = 10$ meters, find the perimeter of the triangle in terms of $\angle BCA$.

$$\sin 28^\circ = \frac{AB}{10} \Rightarrow AB = 10 \sin 28^\circ$$

$$\cos 28^\circ = \frac{AC}{10} \Rightarrow AC = 10 \cos 28^\circ$$

The perimeter is

$$10 + 10 \sin 28^\circ + 10 \cos 28^\circ = 10(1 + \sin 28^\circ + \cos 28^\circ)$$



B. Area of a Triangle

1.33: Measuring the Area of a Triangle in Multiple Ways

The area of a triangle when calculated in different ways, remains the same.

This can be exploited to solve questions that allow the calculation of the area in different ways.

- In other words, the area of a triangle is invariant.
- Invariant simply that it does not vary (or change), based on how you calculate it.

Example 1.34

The altitudes of $\triangle ABC$ are 12, 15, and 20. Find $(\sin \angle A) + (\sin \angle B) + (\sin \angle C)$. (AMC 12P 2002/16, Adapted)

Note: The answer is numerical.

We calculate the area of the triangle in three different ways.

Without loss of generality, let

12 be the altitude associated with base a

15 be the altitude associated with base b

20 be the altitude associated with base c

From the diagram:

$$[ABC] = \frac{1}{2} \cdot 12a = \frac{1}{2} \cdot 15b = \frac{1}{2} \cdot 20c$$

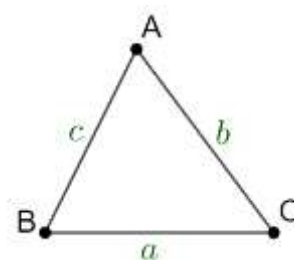
Multiply throughout by 2:

$$2[ABC] = 12a = 15b = 20c$$

Even though the question does not require integer solutions, we treat this as a Diophantine. This is useful because integer solutions are:

- Valid solutions
- Easier to find using number theory method

$$LCM(12, 15, 20) = 60$$



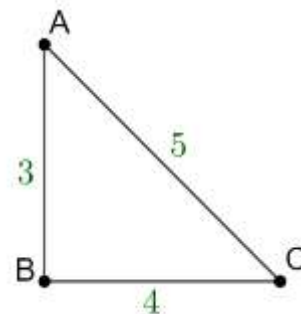
Hence, the above equality has a solution in integers:

$$(a, b, c) = (5, 4, 3)$$

Note that the triangle sides are not (5,4,3), but the sides will be in the ratio:

$(5, 4, 3) \Rightarrow \text{Pythagorean Triplet}$

$$(\sin \angle A) + (\sin \angle B) + (\sin \angle C) = \frac{4}{5} + \frac{5}{5} + \frac{3}{5} = \frac{12}{5}$$



Note: k method for getting solutions for the ratio

$$\begin{aligned} 12a &= 15b = 20c = k \\ a &= \frac{k}{12}, b = \frac{k}{15}, c = \frac{k}{20} \\ a:b:c &= \frac{k}{12} : \frac{k}{15} : \frac{k}{20} = \frac{60}{12} : \frac{60}{15} : \frac{60}{20} = 5:4:3 \end{aligned}$$

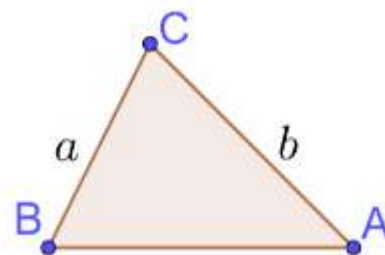
1.35: Area of a Triangle

Area of a triangle with sides a , b and included angle θ is given by:

$$A = \frac{1}{2}ab \sin C$$

Where

a, b are adjacent sides of a triangle
 C is the angle between a and b



Case I: Right Angled Triangle

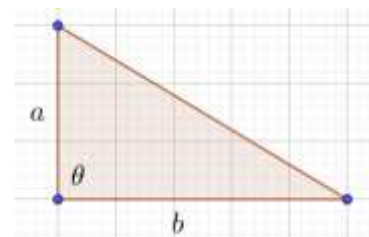
Before we proceed note that, if:

$$C = 90^\circ \Rightarrow \sin C = \sin 90^\circ = 1$$

And the formula becomes:

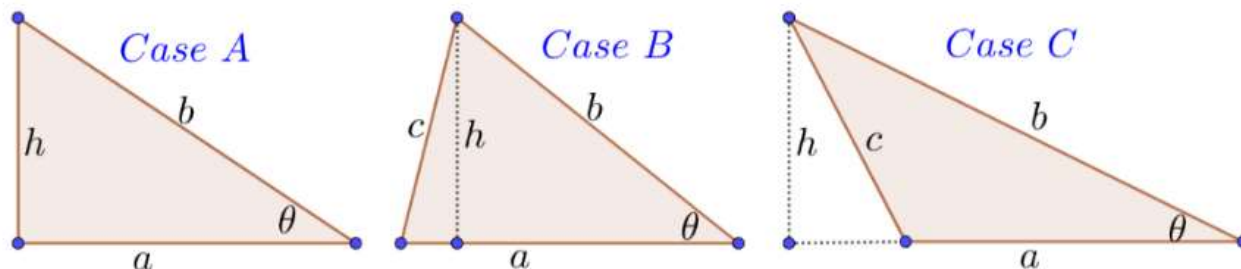
$$A = \frac{1}{2}ab \sin C = \frac{1}{2}ab \cdot 1 = \frac{1}{2}ab$$

Which is the formula for a right-angled triangle that we already know from geometry.



Case II: Three further cases

- One of the angles which is not θ is a right angle.
- Both of the angles which are not θ are acute.
- One of the angles which is not θ is obtuse.



For all three cases, we have:

$$\sin \theta = \frac{\text{opp}}{\text{hyp}} = \frac{h}{b} \Rightarrow h = b \sin \theta$$

Substitute $Height = h = b \sin \theta$, $Base = a$ in $A = \frac{1}{2}hb$:

$$= \frac{1}{2} \underbrace{(b \sin \theta)}_{\text{Height}} \underbrace{(a)}_{\text{Base}} = \frac{1}{2}ab \sin \theta$$

Example 1.36: Triangles

- The adjacent sides of a triangle are 3 feet and 4 feet long. The angle between the two sides is 45° . Find the area of the triangle.
- Two sides of a triangular field are 5 m and 7m long, and the angle between them is 30° . Find the area of the field.
- Anshuman is playing in the garden. He runs 7 feet north, makes a sharp turn at a 60° angle from north, and runs approximately northeast another 10 feet. If he now runs back to his starting position, what is the area in feet of the enclosed figure that he has run around.
- A triangle has sides of length 2 cm and 3 cm. The angles of the triangles are in the ratio 3: 2: 1. What is the sum of the possible values of the area of the triangle.

Part A

$$A = \frac{1}{2}ab \sin 45^\circ = \frac{1}{2} \cdot 3 \cdot 4 \cdot \frac{\sqrt{2}}{2} = 3\sqrt{2}$$

Part B

$$A = \frac{1}{2}ab \sin 30^\circ = \frac{1}{2} \cdot 5 \cdot 7 \cdot \frac{1}{2} = \frac{35}{4} = 8\frac{3}{4}$$

Part C

$$A = \frac{1}{2}ab \sin 120^\circ = \frac{1}{2} \cdot 7 \cdot 10 \cdot \frac{\sqrt{3}}{2} = \frac{35\sqrt{3}}{2}$$

Part D

Let the angles be:

$$3: 2: 1 = 3x: 2x: x$$

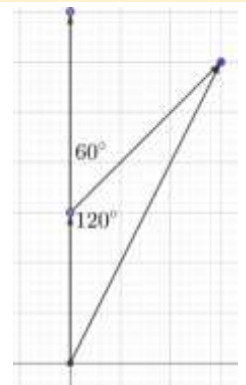
$$3x + 2x + x = 180 \Rightarrow 6x = 180 \Rightarrow x = 30$$

The angles of the triangle are

$$\{30, 60, 90\}$$

The three possibilities are:

$$\begin{aligned} & \frac{1}{2}ab \sin 30^\circ + \frac{1}{2}ab \sin 60^\circ + \frac{1}{2}ab \sin 90^\circ \\ &= \frac{1}{2}ab(\sin 30^\circ + \sin 60^\circ + \sin 90^\circ) \\ &= \frac{1}{2} \cdot 2 \cdot 3 \left(\frac{1}{2} + \frac{\sqrt{3}}{2} + 1 \right) \\ &= 3 \left(\frac{\sqrt{3} + 3}{2} \right) \\ &= \frac{3\sqrt{3} + 9}{2} \end{aligned}$$



1.37: Splitting a Shape

The area of two similar triangles with scale factor k is in the ratio k^2 .

Consider $\triangle ABC \sim \triangle XYZ$.

The ratio of areas of the two triangles is:

$$\frac{[ABC]}{[XYZ]} = \frac{\frac{1}{2} \cdot AB \cdot BC \cdot \sin B}{\frac{1}{2} \cdot XY \cdot YZ \cdot \sin Y}$$

Equation I

But since the triangles are similar:

$$\frac{AB}{XY} = \frac{BC}{YZ} = \frac{AC}{XZ} = k \Rightarrow AB = k \cdot XY, BC = k \cdot YZ, \sin B = \sin Y$$

Make these substitutions in Equation I:

$$\frac{[ABC]}{[XYZ]} = \frac{\frac{1}{2} \cdot (k \cdot XY) \cdot (k \cdot YZ) \cdot \sin Y}{\frac{1}{2} \cdot XY \cdot YZ \cdot \sin Y} = k^2$$

C. Reducing Quadrilaterals

1.38: Splitting a Shape

The area of a shape can be calculated by splitting into smaller shapes that are easier to handle.

Example 1.39

In quadrilateral $ABCD$, $m\angle B = m\angle C = 120^\circ$, $AB = 3$, $BC = 4$, and $CD = 5$. Find the area of $ABCD$. (AMC 12P 2002/19)

Draw a diagram. Construct:

$$AE \parallel BC, \quad BF \perp AE, \quad CG \perp AE$$

Note that

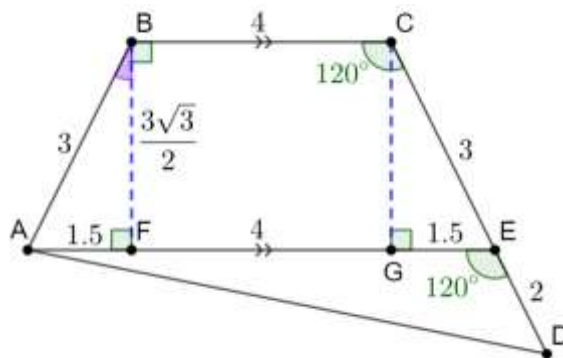
$$\angle ABF = 120 - \angle FBC = 120 - 90 = 30^\circ \Rightarrow \triangle ABF \text{ is } 30 - 60 - 90 \text{ triangle}$$

Hence:

$$AF = \frac{AB}{2} = \frac{3}{2}, BF = AF \cdot \frac{\sqrt{3}}{2} = \frac{3\sqrt{3}}{2}$$

Also, $\triangle CGE \cong \triangle ABF$

$$GE = \frac{3}{2}$$



The area of trapezoid $ABCE$

$$= \frac{b_1 + b_2}{2} \cdot h = \frac{BC + AE}{2} \cdot BF = \frac{4 + (4 + 1.5 + 1.5)}{2} \cdot \frac{3\sqrt{3}}{2} = \frac{4 + 7}{2} \cdot \frac{3\sqrt{3}}{2} = \frac{33\sqrt{3}}{4}$$

The area of $\triangle AED$

$$= \frac{1}{2} ab \sin \theta = \frac{1}{2} (AE)(ED) \sin 120^\circ = \frac{1}{2} (7)(2) \left(\frac{\sqrt{3}}{2} \right) = \frac{14\sqrt{3}}{4}$$

Hence, the final answer is:

$$[ABCE] + [AED] = \frac{33\sqrt{3}}{4} + \frac{14\sqrt{3}}{4} = \frac{47\sqrt{3}}{4}$$

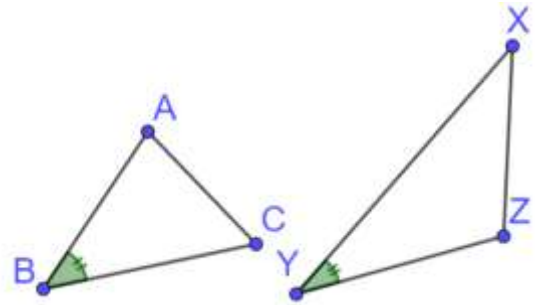
D. Applications

1.40: Triangles with Common Angles

If $\triangle ABC$ and $\triangle XYZ$ have $\angle B = \angle Y$, then the ratio of their areas is given by the ratio of the respective product of the included sides:

$$\frac{[ABC]}{[XYZ]} = \frac{AB \cdot BC}{YX \cdot YZ}$$

$$\frac{[ABC]}{[XYZ]} = \frac{\frac{1}{2} AB \cdot BC \cdot \sin \angle B}{\frac{1}{2} XY \cdot YZ \cdot \sin \angle Y} = \frac{AB \cdot BC}{XY \cdot YZ}$$



1 Pending

Example 1.41

If an angle of a triangle remains unchanged but each of its two including sides is doubled, then the area is multiplied by: (AHSME 1956/27)

Let the sides of the original triangle be

a and b

Then the new sides will be

$2a$ and $2b$

The ratio is then

$$\frac{2a \cdot 2b}{a \cdot b} = 4$$

2 Pending

Example 1.42

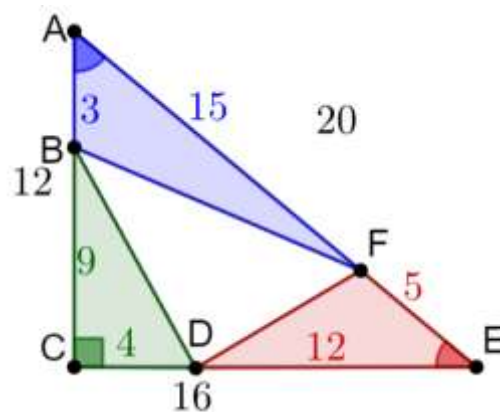
In the right triangle $\triangle ACE$, we have $AC = 12$, $CE = 16$, and $EA = 20$. Points B , D , and F are located on AC , CE , and EA , respectively, so that $AB = 3$, $CD = 4$, and $EF = 5$. What is the ratio of the area of $\triangle DBF$ to that of $\triangle ACE$? (AMC 10B 2004/18)

$$\frac{[ABF]}{[ACE]} = \frac{AB \times AF}{AC \times AE} = \frac{3 \times 15}{12 \times 20} = \frac{3}{16}$$

$$\begin{aligned}\frac{[BCD]}{[ACE]} &= \frac{BC \times CD}{AC \times CE} = \frac{9 \times 4}{12 \times 16} = \frac{3}{16} \\ \frac{[FED]}{[ACE]} &= \frac{EF \times ED}{AE \times EC} = \frac{5 \times 12}{20 \times 16} = \frac{3}{16}\end{aligned}$$

Using Complementary Areas:

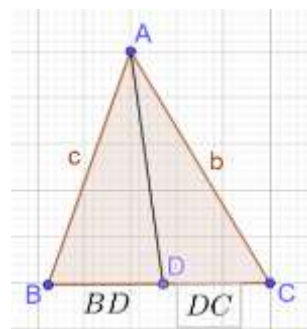
$$\begin{aligned}\frac{[DBF]}{[ACE]} &= 1 - \left(\frac{[ABF]}{[ACE]} + \frac{[BCD]}{[ACE]} + \frac{[FED]}{[ACE]} \right) = 1 - \left(\frac{3}{16} + \frac{3}{16} + \frac{3}{16} \right) \\ &= 1 - \frac{9}{16} = \frac{7}{16}\end{aligned}$$



1.43: Angle Bisector Theorem

An angle bisector of a triangle divides the side opposite it in the ratio of the other two sides of the triangle.

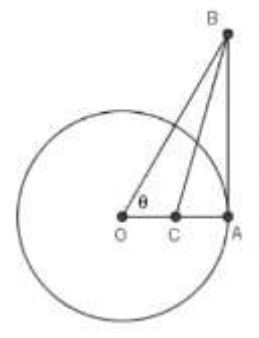
$$\frac{BD}{AB} = \frac{DC}{AC}$$



Example 1.44

A circle centered at O has radius 1 and contains the point A . The segment AB is tangent to the circle at A and $\angle AOB = \theta$. If point C lies on OA and BC bisects $\angle ABO$, then $OC =$ (AMC 12 2000/17)

- A. $\sec^2 \theta - \tan \theta$
- B. $\frac{1}{2}$
- C. $\frac{\cos^2 \theta}{1 + \sin \theta}$
- D. $\frac{1}{1 + \sin \theta}$
- E. $\frac{\sin \theta}{\cos^2 \theta}$



By Angle Bisector Theorem in $\triangle OAB$:

$$\frac{OC}{OB} = \frac{CA}{AB}$$

Equation 1

We need to find the value of OC in terms of θ . Hence, rewrite everything in terms of either OC or in terms of θ .
By tangent perpendicularity theorem:

$$\angle OAB = 90^\circ$$

In right triangle $\triangle OAB$:

$$\begin{aligned}\tan \theta &= \frac{AB}{OA} = \frac{AB}{1} = AB \Rightarrow AB = \tan \theta = \frac{\sin \theta}{\cos \theta} \\ \cos \theta &= \frac{OA}{OB} = \frac{1}{OB} \Rightarrow OB = \frac{1}{\cos \theta} \\ OC + CA &= 1 \Rightarrow AC = 1 - OC\end{aligned}$$

Making the above substitutions in Equation 1:

$$\begin{aligned}\frac{OC}{1} &= \frac{1 - OC}{\frac{\sin \theta}{\cos \theta}} \\ \frac{OC}{1} &= \frac{1 - OC}{\sin \theta} \\ OC \sin \theta &= 1 - OC \\ OC(1 + \sin \theta) &= 1 \\ OC &= \frac{1}{1 + \sin \theta}\end{aligned}$$

Option D

Example 1.45: Law of Sines

We can use the formula for the area of a triangle to prove the law of sines. By considering the area of a triangle in three different ways, show that:

$$\frac{a}{\sin A} = \frac{b}{\sin B} = \frac{c}{\sin C}$$

We calculate the area of the triangle in three different ways:

$$\frac{1}{2}ab \sin C = \frac{1}{2}ac \sin B = \frac{1}{2}bc \sin A$$

Equate the second and the third part:

$$ac \sin B = bc \sin A \Rightarrow \frac{a}{\sin A} = \frac{b}{\sin B}$$

Equation I

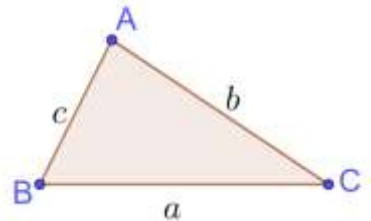
Equate the first and the second part:

$$ab \sin C = ac \sin B \Rightarrow \frac{b}{\sin B} = \frac{c}{\sin C}$$

Equation II

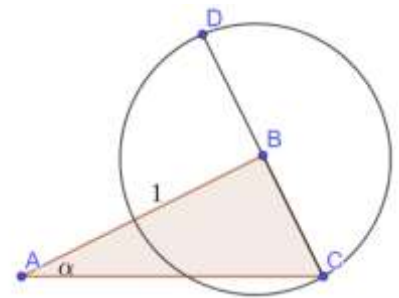
Combine Equation I and II into a tripartite equality:

$$\frac{a}{\sin A} = \frac{b}{\sin B} = \frac{c}{\sin C}$$



Example 1.46

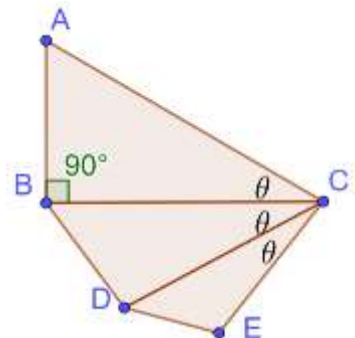
Triangle ABC is a right angle, right angled at B. Side AB has length 1. CD is a diameter of the circle, and B is the center of the circle. $\angle BAC$ is α . Find the diameter of the circle in terms of α .



$$\begin{aligned}\tan \alpha &= \frac{BC}{AB} = \frac{BC}{1} \\ r &= BC = \tan \alpha \\ \text{Diameter} &= 2r = 2 \tan \alpha\end{aligned}$$

Example 1.47

The diagram alongside has three triangles such that $\angle ABC = \angle BDC = \angle DEC = 90^\circ$. Find an expression for DE in terms of AB and θ .



$$\sin \theta = \frac{DE}{DC} \Rightarrow DE = DC \sin \theta$$

$$\begin{aligned}\text{Substitute } \cos \theta = \frac{DC}{BC} \Rightarrow DC &= BC \cos \theta: \\ &= (BC \cos \theta) \sin \theta\end{aligned}$$

$$\begin{aligned}\text{Substitute } \tan \theta = \frac{AB}{BC} \Rightarrow BC &= \frac{AB}{\tan \theta}: \\ &= \frac{AB}{\tan \theta} \cos \theta \sin \theta = AB \cot \theta \cos \theta \sin \theta = AB \cos^2 \theta\end{aligned}$$

Example 1.48

Mark all correct options

Can $\frac{DE}{AB}$ be a:

- A. Positive Integer
- B. Negative Integer
- C. Rational Number
- D. Irrational Number
- E. Integer multiple of the Golden ratio (where golden ratio = $\frac{1+\sqrt{5}}{2}$)

$$\frac{DE}{AB} = \frac{AB \cos^2 \theta}{AB} = \cos^2 \theta$$

$$\cos^2 \theta = 1 \Rightarrow \theta = 0 \Rightarrow \text{Not Valid} \Rightarrow \text{Option A is incorrect}$$

$$\cos^2 \theta < 0 \Rightarrow \text{Not Valid} \Rightarrow \text{Option B is incorrect}$$

If

$$\theta = 60^\circ \Rightarrow \cos \theta = \frac{1}{2} \Rightarrow \cos^2 \theta = \frac{1}{4} \Rightarrow \text{Rational Number} \Rightarrow \text{Option C is correct}$$

$$0 < \cos^2 \theta < 1 \Rightarrow \cos^2 \theta \text{ can be irrational} \Rightarrow \text{Option D is correct}$$

$$\frac{1 + \sqrt{5}}{2} > 1 > \cos^2 \theta \Rightarrow \text{Option E is incorrect}$$

Option C, D

Example 1.49

Finding an interesting expression

Example 1.50: Finding area

In $\triangle ABC$, $\angle ABC = 45^\circ$. Point D is on BC so that $2 \cdot BD = CD$ and $\angle DAB = 15^\circ$. Find $\angle ACB$. (AMC 12 2001/24)

Note: Answer options were all 90° or less.

Since an exterior angle is the sum of two interior angles:

$$\angle ADC = \angle ABD + \angle DAB = 45 + 15 = 60^\circ$$

Draw $AX \perp BC$.

$$\angle DAX = 180 - 90 - 60 = 30^\circ$$

$$\angle BAX = \angle BAD + \angle DAX = 15 + 30 = 45^\circ$$

In Isosceles $\triangle AXB$

$$AX = BX$$

$$AX = BD + DX$$

In $30 - 60 - 90 \triangle ADX$, $DX = \frac{\sqrt{3}}{3} AX$

$$AX = BD + \frac{\sqrt{3}}{3} AX$$

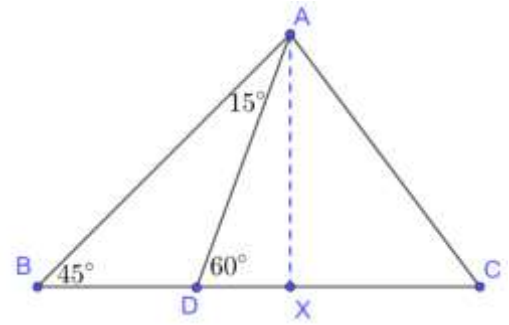
$$AX \left(1 - \frac{\sqrt{3}}{3} \right) = BD$$

$$AX = \frac{BD}{1 - \frac{\sqrt{3}}{3}} = \frac{BD}{\frac{3 - \sqrt{3}}{3}} = \frac{3}{3 - \sqrt{3}} BD = \frac{3 + \sqrt{3}}{2} BD$$

$$XC = CB - XB = 3BD - \frac{3 + \sqrt{3}}{2} BD = BD \left[\frac{6 - 3 - \sqrt{3}}{2} \right] = BD \left[\frac{3 - \sqrt{3}}{2} \right]$$

$$\tan \angle ACX = \frac{AX}{XC} = \frac{\frac{3 + \sqrt{3}}{2} BD}{BD \left[\frac{3 - \sqrt{3}}{2} \right]} = \frac{3 + \sqrt{3}}{3 - \sqrt{3}} = 2 + \sqrt{3}$$

$$\angle ACD = 75^\circ$$



1.4 Polygons

A. Quadrilaterals

1.51: Area of a Parallelogram

The area of a parallelogram is the product of two adjacent sides and the sine of the included angle.

$$s_1 s_2 \sin \angle D$$

Where

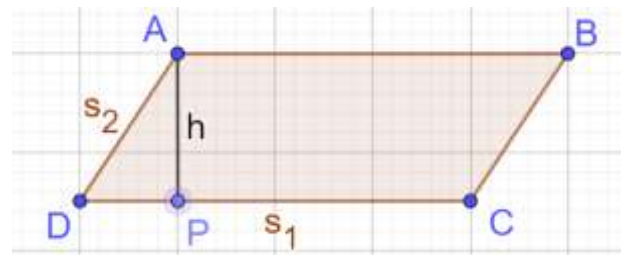
s_1 and s_2 are adjacent sides

D is the included angle

Draw parallelogram $ABCD$ with sides s_1 and s_2 . Draw the height from vertex $A \perp DC$.

In Right Triangle $\triangle ADP$:

$$\sin \angle D = \frac{\text{opp}}{\text{hyp}} = \frac{h}{s_2} \Rightarrow h = \sin \angle D \times s_2$$



Area of Parallelogram

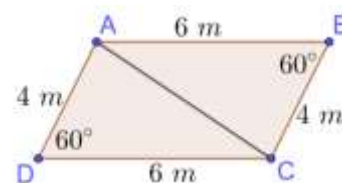
$$= hb = \underbrace{(\sin \angle D \times s_2)}_{\text{Height}} \times \underbrace{s_1}_{\text{Base}} = s_1 s_2 \sin \angle D$$

Example 1.52: Finding area

- Find the area of a parallelogram with sides of $4m$ and $6m$ and an angle between the two sides of 60° .
- A parallelogram has adjacent sides of length 3 cm and 5 cm , and angle between the two sides as 30° . Find the area of the parallelogram.
- A parallelogram has adjacent sides of length 5 cm and 7 cm , and angle between the two sides as 27° . Find a trigonometric expression for the area of the parallelogram.

Part A

$$A = s_1 s_2 \sin \angle D = 4 \times 6 \sin 60^\circ = 4 \times 6 \times \frac{\sqrt{3}}{2} = 12\sqrt{3}$$



Part B

$$A = s_1 s_2 \sin \angle D = 3 \times 5 \times \frac{1}{2} = 7.5$$

Part C

$$A = s_1 s_2 \sin \angle D = 5 \times 7 \times \sin 27^\circ = 35 \sin 27^\circ$$

Example 1.53

- The area of a parallelogram is 1 square meter. The base of the parallelogram is 5 meters. The angle between the base and the adjacent side is 30° . Find the length of the adjacent side.

$$\begin{aligned} \text{Area} &= ab \sin C \\ 1 &= a \times 5 \times \frac{1}{2} \Rightarrow a = \frac{2}{5} m \end{aligned}$$

1.54: Rhombus

- A rhombus is a special case of a parallelogram.
- All sides of a rhombus are equal.

Example 1.55: Rhombus

- A rhombus has a side length of $12m$. An interior angle of the rhombus is 30° . Find the area of the rhombus.
- Find the side length of a rhombus with an area of A and an internal angle of α° .

Part A

$$A = s_1 s_2 \sin \angle D = 12 \cdot 12 \cdot \frac{1}{2} = 72 m^2$$

Part B

$$A = s^2 \times \sin \alpha \Rightarrow s = \sqrt{\frac{A}{\sin \alpha}} \text{ units}$$

1.56: Kites

Example 1.57: Kite

1.58: Area of a General Quadrilateral

Show that the area of a quadrilateral is one half the product of the length of its diagonals and the sine of the angle between the diagonals.

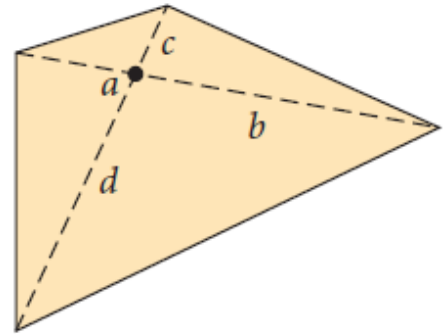
$$\text{Area} = \frac{1}{2} \sin \theta [(a + b)(c + d)]$$

Apply the formula for area of a triangle $A = \frac{1}{2} ab \sin \theta$ to the four triangles that make up the quadrilateral:

$$= \frac{1}{2} ac \sin \theta + \frac{1}{2} ad \sin(180 - \theta) + \frac{1}{2} cb \sin(180 - \theta) + \frac{1}{2} db \sin \theta$$

Substitute $\sin(180 - \theta) = \sin \theta$. This gives a $\frac{1}{2} \sin \theta$ in each term, which we factor out:

$$= \frac{1}{2} \sin \theta [ac + ad + cb + db] = \frac{1}{2} \sin \theta [(a + b)(c + d)]$$



Example 1.59: Area of General Quadrilateral

B. Trigonometry with Rectangles

Example 1.60

A rectangular gate of width w and height h has two cross braces going from the top left to bottom right, and top right to bottom left, respectively. Find the angles between the braces in terms of h and w given that θ is the angle made by the braces with the base.

Since $\triangle ADC \cong \triangle BDC$,

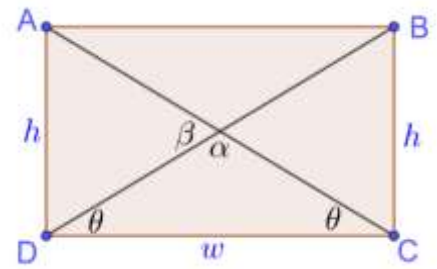
$$\angle BDC = \angle ACD = \theta$$

In $\triangle BDC$:

$$\tan \theta = \frac{\text{opp}}{\text{adj}} = \frac{h}{w} \Rightarrow \theta = \tan^{-1} \frac{h}{w}$$

The angle between the diagonals is

$$\begin{aligned} &= \alpha = 180 - 2\theta = 180 - 2 \left(\tan^{-1} \frac{h}{w} \right) \\ &\beta = 180 - \alpha = 2\theta = 2 \left(\tan^{-1} \frac{h}{w} \right) \end{aligned}$$



Example 1.61

Find the perimeter of a rectangle with diagonal d , and an angle θ between diagonal and side.

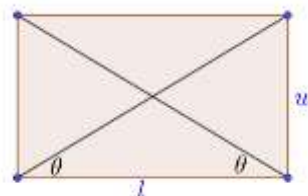
Draw a diagram of a rectangle with width w and length l . Then:

$$\sin \theta = \frac{w}{d} \Rightarrow w = d \sin \theta$$

$$\cos \theta = \frac{l}{d} \Rightarrow l = d \cos \theta$$

Then the perimeter is:

$$P = 2(l + w) = 2(d \sin \theta + d \cos \theta) = 2d(\sin \theta + \cos \theta)$$



Example 1.62

Sandy wants to draw a 2 cm margin for 30 cm long art paper. He doesn't have a ruler, so while he starts 2 cm from the edge of the page, he ends 3 cm away from the edge of the page. Find the angle that the line made by Sandy makes with the parallel to the edge of the paper.

Draw a diagram. We want to find the angle θ .

Method I

$$\tan \theta = \frac{1}{30}$$

Take the tan inverse of both sides:

$$\theta = \tan^{-1}\left(\frac{1}{30}\right)$$

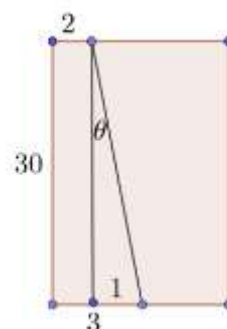
Method II

$$\text{Hyp} = \sqrt{901}$$

$$\frac{1}{2} \cdot 30 \cdot \sqrt{901} \sin \theta = 15$$

$$\sin \theta = \frac{1}{\sqrt{901}}$$

$$\theta = \sin^{-1}\left(\frac{1}{\sqrt{901}}\right)$$



Example 1.63

A rectangular piece of paper is divided into a square and a rectangle. The rectangle is folded along the diagonal to create a triangle. Show that the ratio of the perimeter of the triangle to the perimeter of the square is independent of the length of the diagonal.

The ratio is:

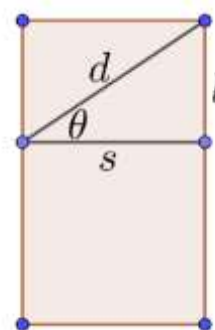
$$\frac{P(\text{Triangle})}{P(\text{Square})} = \frac{d + s + l}{4s}$$

This ratio has d in it, which we would like to eliminate.

Substitute $\cos \theta = \frac{s}{d} \Rightarrow s = d \cos \theta$ and $\sin \theta = \frac{l}{d} \Rightarrow l = d \sin \theta$:

$$\frac{d + d \cos \theta + d \sin \theta}{4d \cos \theta} = \frac{1 + \sin \theta + \cos \theta}{4 \cos \theta}$$

The last expression does not contain d . Hence, it is independent of d .



1.64: Double Angle Identity

$$\sin 2x = 2 \sin x \cos x$$

Challenge 1.65

In a rectangle $ABCD$, E is the midpoint of AB . F is a point in AC such that BF is perpendicular to AC , and FE is perpendicular to BD . Suppose $BC = 8\sqrt{3}$. Find AB . (IOQM 2017/13)

Angle Chasing

Let $\angle DBA = \angle CAB = \alpha$.

In $\triangle ENB$

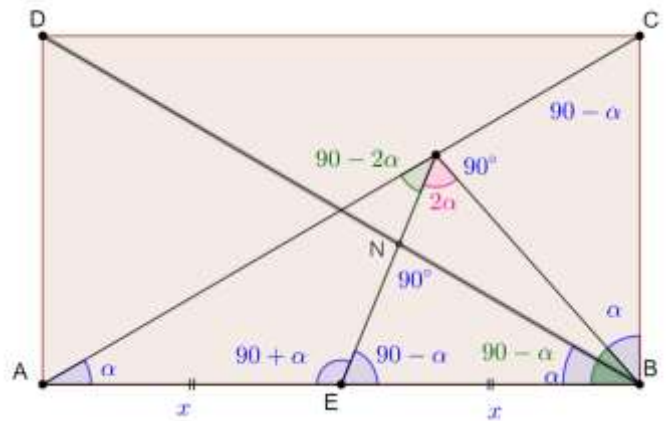
$$\begin{aligned}\angle NEB &= 90 - \alpha \\ \angle FEA &= 180 - (90 - \alpha) = 90 + \alpha\end{aligned}$$

In $\triangle AFE$:

$$\angle AFE = 180 - \alpha - (90 + \alpha) = 90 - 2\alpha$$

In $\triangle AFB$

$$\begin{aligned}\angle AFB &= 90^\circ \quad (BF \perp AC) \\ \angle EFB &= 90 - (90 - 2\alpha) = 2\alpha\end{aligned}$$



Right Triangle Trigonometry

Let $AE = EB = x$.

$$\text{In right } \triangle ENB: \cos \alpha = \frac{BN}{x} \Rightarrow \underline{BN = x \cos \alpha} \quad \text{Equation I}$$

$$\text{In right } \triangle AFB: \sin \alpha = \frac{BF}{2x} \Rightarrow BF = 2x \sin \alpha$$

$$\text{In right } \triangle BFN: \sin 2\alpha = \frac{BN}{BF} \Rightarrow BN = BF \sin 2\alpha \Rightarrow \underline{BN = (2x \sin \alpha)(\sin 2\alpha)} \quad \text{Equation II}$$

$$\begin{aligned}x \cos \alpha &= (2x \sin \alpha)(\sin 2\alpha) \\ \cos \alpha &= (2 \sin \alpha)(2 \sin \alpha \cos \alpha) \\ \frac{1}{4} &= \sin^2 \alpha \Rightarrow \sin \alpha = \frac{1}{2} \Rightarrow \alpha = 30^\circ\end{aligned}$$

Special Right Triangles

$$\tan 30^\circ = \frac{BC}{AB} \Rightarrow AB = \frac{BC}{\tan 30^\circ} = \frac{8\sqrt{3}}{\frac{1}{\sqrt{3}}} = (8\sqrt{3})\sqrt{3} = 24$$

C. Regular Polygons

Example 1.66

In regular hexagon $ABCDEF$, what is $\angle ADE$?

All regular hexagons are similar. Without loss of generality, consider a regular hexagon with side length 1.

Each angle of a regular hexagon is:

$$= \frac{(n-2)(180)}{n} = \frac{(6-2)(180)}{6} = 120$$

Draw $AE \perp ED$.

$$\angle AEF = \underbrace{120}_{\angle FED} - 90 = 30^\circ$$

Draw $FM \perp AE$

$$\angle MFE = 180 - 90 - 30 = 60^\circ$$

$$\sin \angle MFE = \frac{ME}{FE} \Rightarrow ME = FE \cdot \sin 60^\circ = 1 \cdot \frac{\sqrt{3}}{2} = \frac{\sqrt{3}}{2}$$

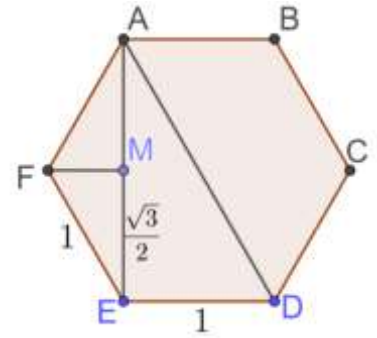
$$AE = 2 \cdot ME = 2 \times \frac{\sqrt{3}}{2} = \sqrt{3}$$

$$\tan \angle ADE = \frac{AE}{ED} = \sqrt{3} \Rightarrow \angle ADE = 60^\circ$$

Method II

Recognize that

$$\begin{aligned} AD &= 2 \\ \cos \angle ADE &= \frac{1}{2} \Rightarrow \angle ADE = 60^\circ \end{aligned}$$



1.67: Apothem

The apothem of a regular polygon is the line segment from the center to the midpoint of one of its side.

Example 1.68

Area of a regular hexagon

1.69: Some Trig Values

$$\tan 36^\circ = \sqrt{5 - 2\sqrt{5}}$$

$$\tan 22.5^\circ = \sqrt{2} - 1$$

$$\tan 15^\circ = 2 - \sqrt{3}$$

Example 1.70

Find the area of a regular octagon with side s .

By connecting the vertices, we get 8 congruent triangles. The angle subtended by each triangle is $\frac{1}{8}$ of a circle.

$$\angle AOB = \frac{360}{8} = 45$$

Draw $OP \perp AB$

$$\angle POB = \theta = \angle \frac{AOB}{2} = \frac{45}{2} = 22.5^\circ$$

Also, note that:

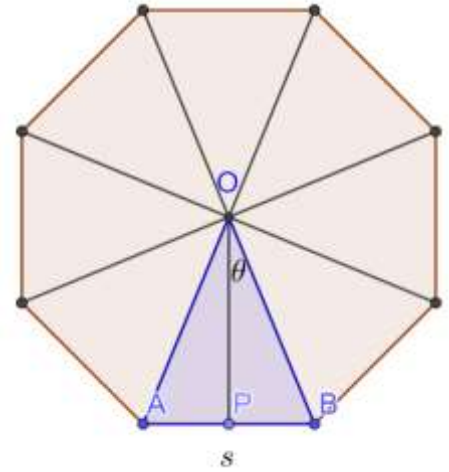
$$\tan \theta = \frac{PB}{OP} \Rightarrow OP = \frac{PB}{\tan \theta} = \frac{PB}{\tan 22.5^\circ} = \frac{PB}{\sqrt{2} - 1}$$

Substitute $AB = s \Rightarrow PB = \frac{s}{2}$:

$$\frac{\frac{s}{2}}{\sqrt{2} - 1} = \frac{s}{2(\sqrt{2} - 1)} \cdot \frac{\sqrt{2} + 1}{\sqrt{2} + 1} = \frac{s(\sqrt{2} + 1)}{2(2 - 1)} = \frac{s(\sqrt{2} + 1)}{2}$$

The area of the octagon is eight times the area of area of $\triangle APB$:

$$= \underbrace{\frac{1}{2} \times s \times \frac{s(\sqrt{2} + 1)}{2}}_{\text{Area of } \triangle APB} \times 8 = 2s^2(\sqrt{2} + 1)$$



Example 1.71

Find the area of a regular dodecagon with side s .

By connecting the vertices, get 12 congruent triangles. The angle subtended by each triangle is $\frac{1}{12}$ of a circle.

$$\angle AOB = \frac{360}{12} = 30$$

Draw $OP \perp AB$

$$\angle POB = \theta = \angle \frac{AOB}{2} = \frac{30}{2} = 15^\circ$$

Note that:

$$\tan \theta = \frac{PB}{OP} \Rightarrow OP = \frac{PB}{\tan \theta} = \frac{PB}{\tan 15^\circ} = \frac{PB}{2 - \sqrt{3}}$$

Substitute $AB = s \Rightarrow PB = \frac{s}{2}$:

$$\frac{\frac{s}{2}}{2 - \sqrt{3}} = \frac{s}{2(2 - \sqrt{3})} \cdot \frac{2 + \sqrt{3}}{2 + \sqrt{3}} = \frac{s(2 + \sqrt{3})}{2(4 - 3)} = \frac{s(2 + \sqrt{3})}{2}$$

The area of the dodecagon is twelve times the area of $\triangle APB$:

$$= \frac{1}{2} \times s \times \frac{s(2 + \sqrt{3})}{2} \times 12 = 3s^2(2 + \sqrt{3})$$

Example 1.72

Find the area of a regular pentagon with side s .

By connecting the vertices, get 5 congruent triangles. The angle subtended by each triangle is $\frac{1}{5}$ of a circle.

Draw $OP \perp AB$.

$$\angle POB = \theta = \frac{360}{10} = 36^\circ$$

Also, note that:

$$\tan \theta = \frac{PB}{OP} \Rightarrow OP = \frac{PB}{\tan \theta} = \frac{PB}{\tan 36^\circ} = \frac{PB}{\sqrt{5-2\sqrt{5}}}$$

Substitute $AB = s \Rightarrow PB = \frac{s}{2}$, and rationalize:

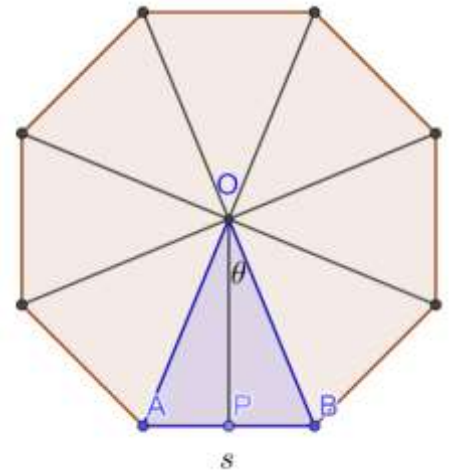
$$\frac{\frac{s}{2}}{\sqrt{5-2\sqrt{5}}} = \frac{s}{2\sqrt{5-2\sqrt{5}}} \cdot \frac{\sqrt{5+2\sqrt{5}}}{\sqrt{5+2\sqrt{5}}} = \frac{s(\sqrt{5+2\sqrt{5}})}{2\sqrt{5}}$$

Rationalize one more time:

$$= \frac{s(\sqrt{5+2\sqrt{5}})}{2\sqrt{5}} \cdot \frac{\sqrt{5}}{\sqrt{5}} = \frac{s(\sqrt{5(5+2\sqrt{5})})}{10}$$

The area of the pentagon is five times the area of $\triangle APB$:

$$= \frac{1}{2} \times s \times \frac{s(\sqrt{5(5+2\sqrt{5})})}{10} \times 5 = \frac{1}{4} s^2 (\sqrt{5(5+2\sqrt{5})})$$



1.73: Regular Polygons: General Formula

$$\frac{1}{2} \times \text{Perimeter} \times \text{Apothem}$$

D. Regular Polygons and the Circle

Example 1.74

1962 AHSME Problems/Problem 18

Problem

A regular dodecagon (12 sides) is inscribed in a circle with radius r inches. The area of the dodecagon, in square inches, is:

Divide the dodecagon into 12 congruent triangles, each of which subtends an angle at the center of:

$$\frac{360^\circ}{12} = 30^\circ$$

Each of the triangles has area:

$$\frac{1}{2}ab \sin \theta = \frac{1}{2}r^2 \sin 30^\circ = \frac{1}{2}r^2 \left(\frac{1}{2}\right) = \frac{r^2}{4}$$

And the area of the entire polygon will be:

$$12 \cdot \frac{r^2}{4} = 3r^2$$

Example 1.75

A circle with radius r has a regular polygon inscribed in it. Show that the area of the polygon is:

$$\frac{1}{2}r^2 n \sin\left(\frac{2\pi}{n}\right)$$

Consider a regular polygon inscribed in a circle. For example, the diagram has an octagon inscribed in a circle. Draw the radii from the center of the circle to the vertices of the polygon.

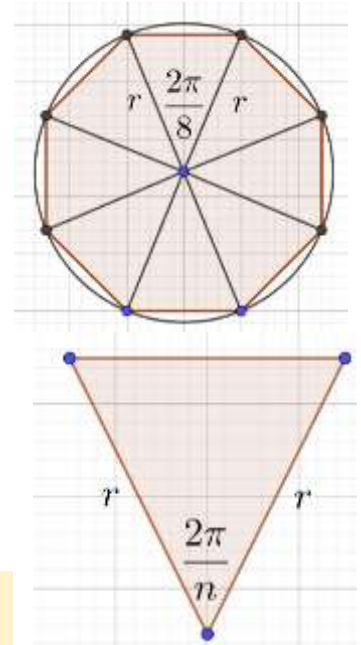
The radii are equal, and the angle subtended by each triangle is equal, since it is a regular polygon. Hence, by SAS, the triangles are all congruent.

The area of a single triangle

$$= \frac{1}{2}ab \sin \theta = \frac{1}{2}r^2 \sin\left(\frac{2\pi}{n}\right)$$

The area of the entire polygon

$$= \underbrace{n}_{\text{No. of Triangles}} \times \underbrace{\frac{1}{2}r^2 \sin\left(\frac{2\pi}{n}\right)}_{\text{Area of a Triangle}} = \frac{1}{2}r^2 n \sin\left(\frac{2\pi}{n}\right)$$



Example 1.76

A circle with radius r has a regular polygon circumscribing it. Show that the area of the polygon is:

$$r^2 n \tan \frac{\pi}{n}$$

Consider a regular polygon circumscribed around a circle. For example, the diagram has a hexagon circumscribed around a circle. Draw the radii from the center of the circle to the vertices of the polygon.

Note the radii are all equal, and the angle subtended by each triangle is equal, since it is a regular polygon. Hence, by SAS Theorem, the triangles are all congruent.

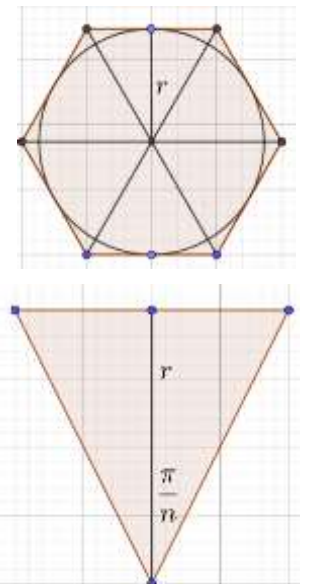
$$\tan \frac{\pi}{n} = \frac{\frac{1}{2} \text{Base}}{r} \Rightarrow \text{Base} = 2r \tan \frac{\pi}{n}$$

The area of a single triangle

$$= \frac{1}{2} \times \underbrace{r}_{\text{Height}} \times \underbrace{2r \tan \frac{\pi}{n}}_{\text{Base}} = r^2 \tan \frac{\pi}{n}$$

The area of the polygon

$$= \underbrace{n}_{\text{No. of Triangles}} \times \underbrace{r^2 \tan \frac{\pi}{n}}_{\text{Area of a Triangle}} = r^2 n \tan \frac{\pi}{n}$$



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Aziz Manva (azizmanva@gmail.com)

2. HEIGHTS AND DISTANCES

2.1 Heights and Distances: Basics

A. Angle of Elevation and Depression

2.1: Angle of Elevation and Depression

- **Angle of Elevation:** Angle between the horizontal and a line of sight above the horizontal
- **Angle of Depression:** Angle between the horizontal and a line of sight below the horizontal

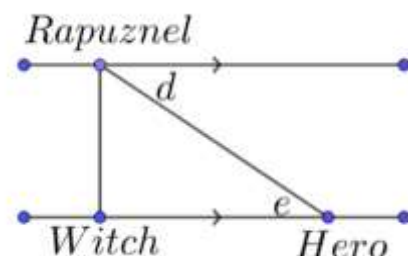
The straight line from Rapunzel to the Hero is a transversal cutting the two parallel horizontal lines. Hence:

d and e are alternate interior angles

They are $\cong$

$$\underbrace{d}_{\text{Angle of Depression}} = \underbrace{e}_{\text{Angle of Elevation}}$$

We are interested in using triangles to solve real life problems. The definitions above are used in the applications below.



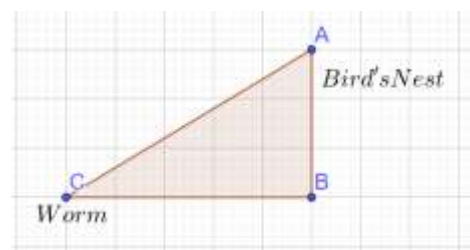
(Calc) Example 2.2

A bird sitting in its nest on a tree observes a worm at an angle of depression of 32° . The bird flies a distance of 23 meters to the worm and back to its nest. Find the:

- height of the nest from the ground.
- ground distance between the worm and the tree

$$\sin 32^\circ = \frac{AB}{23} \Rightarrow AB = \frac{23}{2} \sin 32^\circ = 6.09 \text{ m}$$

$$\cos 32^\circ = \frac{BC}{23} \Rightarrow BC = \frac{23}{2} \cos 32^\circ = 9.75 \text{ m}$$



Example 2.3

An observation station observes an angle of elevation of β to a balloon that was released from x miles away. Assuming the balloon rose in a straight line, determine the angle of depression from the balloon to the observation station. Then, find the height h reached by the balloon, and the distance d from the observation station to the balloon.

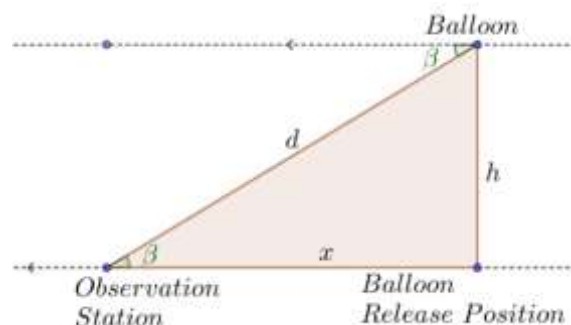
Draw a line on the ground, and another parallel to it at the height of the balloon. By alternate interior angles:

$$\text{Angle of depression} = \text{Angle of Elevation} = \beta$$

Find the height and the distance:

$$\tan \beta = \frac{h}{x} \Rightarrow h = x \tan \beta$$

$$\cos \beta = \frac{x}{d} \Rightarrow d = \frac{x}{\cos \beta}$$



Example 2.4

A boat floating on the water observes the light at the top of the lighthouse of height h at an angle of elevation of α . The horizontal distance of the boat from the cliff that has the lighthouse at its edge is d . Find the height of the cliff.

Let the height of the cliff be c .

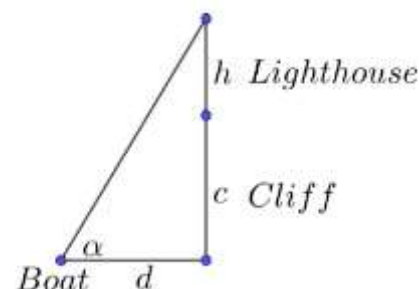
$$\tan \alpha = \frac{h + c}{d}$$

Clearing fractions:

$$h + c = d \tan \alpha$$

Solving for c :

$$c = d \tan \alpha - h$$



Example 2.5

A ladder l units long makes an angle γ with the ground when resting against the window of a wall at a horizontal distance d units away. Show that $\gamma = \tan^{-1} \left(\frac{\sqrt{(l+d)(l-d)}}{d} \right)$.

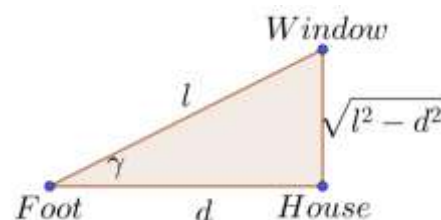
By the Pythagorean Theorem,

$$\text{height of wall} = \sqrt{l^2 - d^2}$$

$$\tan \gamma = \frac{\sqrt{l^2 - d^2}}{d}$$

Solving for γ :

$$\gamma = \tan^{-1} \left(\frac{\sqrt{l^2 - d^2}}{d} \right) = \tan^{-1} \left(\frac{\sqrt{(l+d)(l-d)}}{d} \right)$$



Example 2.6

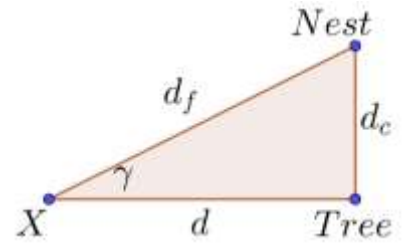
A bird finds a shiny toy at point X, and flies back to its nest on a tree, flying at an angle of γ to the ground and travelling a distance of d_f . A boy at point X runs d units after the bird, and then climbs d_c units to get his toy back. Show that $d_{fly} - d_c = \frac{d(1 - \sin \gamma)}{\cos \gamma}$.

$$\tan \gamma = \frac{d_c}{d} \Rightarrow d_c = d \tan \gamma$$

$$\cos \gamma = \frac{d}{d_f} \Rightarrow d_f = \frac{d}{\cos \gamma}$$

Substitute the above two into $d_f - d_c$:

$$\frac{d}{\cos \gamma} - d \tan \gamma = \frac{d(1 - \sin \gamma)}{\cos \gamma}$$



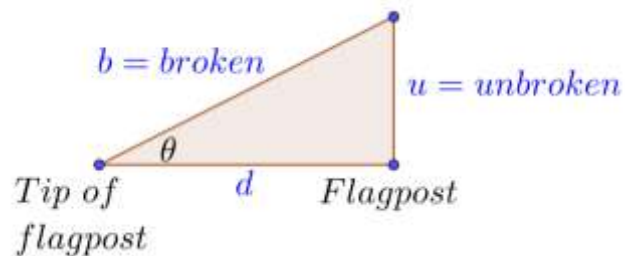
Example 2.7

A flag post breaks at a point and touches the ground d units away from its base. If the angle made by the broken part of the flag post with the ground is θ , then find the length of the flag post before it was broken.

$$\tan \theta = \frac{u}{d} \Rightarrow u = d \tan \theta$$

$$\cos \theta = \frac{d}{b} \Rightarrow b = \frac{d}{\cos \theta}$$

$$u + b = d \tan \theta + \frac{d}{\cos \theta} = d \left(\frac{1 + \sin \alpha}{\cos \alpha} \right)$$



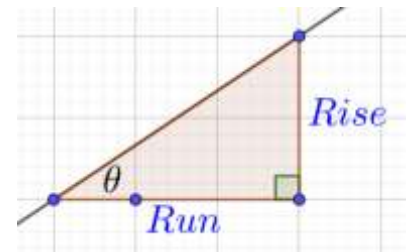
B. $\tan \theta$ as Slope

2.8: $\tan \theta$ as Slope

$$\text{Slope} = \tan \theta$$

$$\text{Slope} = \frac{y_2 - y_1}{x_2 - x_1} = \frac{\text{Rise}}{\text{Run}}$$

$$\tan \theta = \frac{\text{Rise}}{\text{Run}}$$



Example 2.9: Railway Engineering

The Manitou and Pike's Peak Railway goes from its base station in Colorado (elevation c feet) to Pikes Peak (elevation p feet) of the Rocky Mountains. The horizontal distance between the base station and Pike's peak is d feet. Find the angle of elevation from the base station to Pikes Peak

- in terms of the $\tan$ inverse function
- in terms of the $\sin$ inverse function

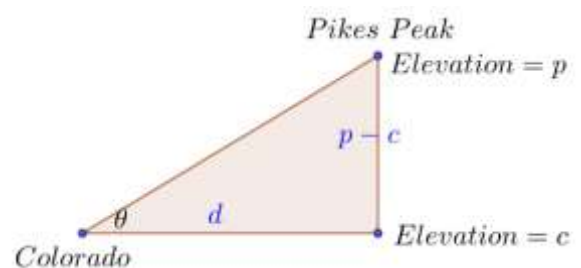
Part A

$$\tan \theta = \frac{p - c}{d} \Rightarrow \theta = \tan^{-1} \left(\frac{p - c}{d} \right)$$

Part B

$$\text{Hyp}^2 = (p - c)^2 + d^2$$

$$\sin \theta = \frac{p - c}{\sqrt{(p - c)^2 + d^2}} \Rightarrow \theta = \sin^{-1} \left(\frac{p - c}{\sqrt{(p - c)^2 + d^2}} \right)$$



Example 2.10: Railway Engineering

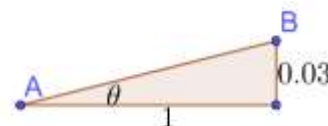
From coordinate geometry, recall that $\tan \theta$ represents the slope of a line when the triangle is set up appropriately. In railway engineering, the *grade of a track* is the slope of a railroad track. It can be expressed in various ways, including as a percentage of its rise for the length of its run. Find the angle of elevation for a

train going uphill at a grade of 3%.

Let the train start at point A, and go till point B. We can take any length that we want for the horizontal distance travelled. Let the distance be 1.

Then:

$$\tan \theta = 0.03 \Rightarrow \theta = \tan^{-1} 0.03$$



Example 2.11

A rise of 600 feet is required to get a railroad line over a mountain. The grade can be kept down by lengthening the track and curving it around the mountain peak. The additional length of track required to reduce the grade from 3% to 2% is approximately (answer to one significant digit): (AHSME 1951/38)

Calculate horizontal distance required

For a 2% grade

$$600 = 2\% \text{ of } AB \Rightarrow AB = \frac{600}{2\%} = 600 \times \frac{100}{2} = 30,000$$

For a 3% grade

$$600 = 3\% \text{ of } AB \Rightarrow AB = \frac{600}{3\%} = 600 \times \frac{100}{3} = 20,000$$



Difference

$$= 30,000 - 20,000 = 10,000$$

Approximate to get actual track length

The actual track length

$$= AC$$

In this context:

$$AC \approx 10,000$$

C. Two Triangles: Horizontal Movement

Another way to increase the complexity of a situation is through having more than one triangle where the trigonometric ratios need to be manipulated. The standard question of this type often requires calculating a quantity in two different ways, and then equating them. (This is equivalent to solving a system of equations).

Example 2.12

A person standing on the bank of a river observes that the angle of elevation of the top of a tree on the opposite bank of the river is 60° , and when he retires 40 meters away from the tree the angle of elevation becomes 30° . The breadth of the river is: (JEE Main 2004)

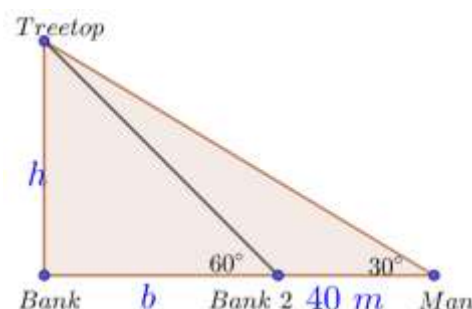
Draw a diagram. Let

$$\begin{aligned} \text{Height of tree} &= h \\ \text{Breadth of river} &= b \end{aligned}$$

Find the height of the tree using the triangle with the greater angle of elevation:

$$\tan 60^\circ = \frac{h}{b} \Rightarrow h = b \tan 60^\circ = \sqrt{3}b$$

Equation I



Find the height of the tree using the triangle with the smaller angle of elevation:

$$\tan 30^\circ = \frac{h}{b+40} \Rightarrow h = \frac{b+40}{\sqrt{3}}$$

Equation II

But the height in both the cases must be same.

Compare Equations I and II and set the RHS equal:

$$\sqrt{3}b = \frac{b+40}{\sqrt{3}}$$

Cross-multiply and solve the resulting linear equation:

$$3b = b + 40 \Rightarrow b = 20$$

Example 2.13: General Case

The angle of inclination to a mountain of height h is θ_1 . After moving a distance d closer to the mountain, the angle increases to θ_2 , with the point on the ground below the mountain peak x units away. Show that

$$h = d \frac{\tan \theta_2 \tan \theta_1}{\tan \theta_2 - \tan \theta_1}$$

In the right triangle with angle θ_2 :

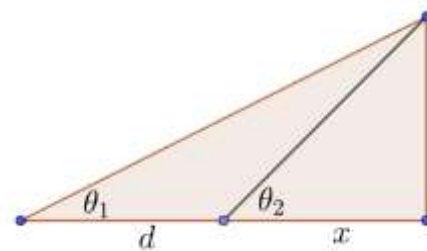
$$\tan \theta_2 = \frac{h}{x} \Rightarrow x = \frac{h}{\tan \theta_2}$$

Equation I

In the right triangle with angle θ_1 :

$$\tan \theta_1 = \frac{h}{d+x} \Rightarrow \tan \theta_1 d + \tan \theta_1 x = h \Rightarrow x = \frac{h - \tan \theta_1 d}{\tan \theta_1}$$

Equation II



Equate the RHS of Equations I and II:

$$\frac{h}{\tan \theta_2} = \frac{h - \tan \theta_1 d}{\tan \theta_1}$$

Cross-multiply:

$$h \tan \theta_1 = h \tan \theta_2 - \tan \theta_1 \tan \theta_2 d$$

Collate all h terms on the RHS, factor out h :

$$\tan \theta_2 \tan \theta_1 d = h(\tan \theta_2 - \tan \theta_1)$$

Solve for h :

$$h = d \cdot \frac{\tan \theta_2 \tan \theta_1}{\tan \theta_2 - \tan \theta_1}$$

Example 2.14

AB is a vertical pole with B at the ground level and A at the top. A man finds that the angle of elevation from a certain point C on the ground is 60° . He moves away from the pole along the line BC to a point D such that $CD = 7m$. From D , the angle of elevation of the point A is 45° . Then the height of the pole is: (JEE Main 2008)

Note: Rationalize the denominator of the answer

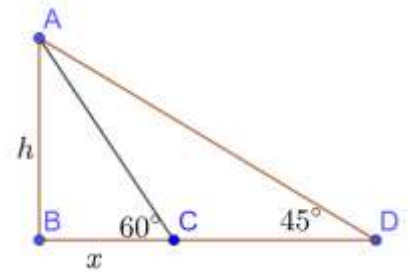
$$h = d \frac{\tan \theta_1 \tan \theta_2}{\tan \theta_2 - \tan \theta_1}$$

Substitute $d = 7$, $\tan \theta_1 = \tan 45^\circ = 1$, $\tan \theta_2 = \tan 60^\circ = \sqrt{3}$:

$$= 7 \frac{(1)(\sqrt{3})}{\sqrt{3} - 1}$$

Rationalize the denominator:

$$= \frac{7\sqrt{3}(\sqrt{3} + 1)}{2} = \frac{21 + 7\sqrt{3}}{2}$$



2.15: Double Angle Identity for $\tan$

$$\tan 2\theta = \frac{2 \tan \theta}{1 - \tan^2 \theta}$$

Example 2.16

The central pole of a conical tent is $\frac{3}{2}$ meter high. The pole is supported by ropes tied to its top and nails on the ground. If, on the ground from the foot of the pole, the distances of the surface of the tent and the nail(s) are in the ratio of 1:3, and if the angles of depression from the top of the pole to the nails, and the surface of the tent are in the ratio of 1:2, then the length of one such rope is: (JMET 2011/78)

From the rope and the pole:

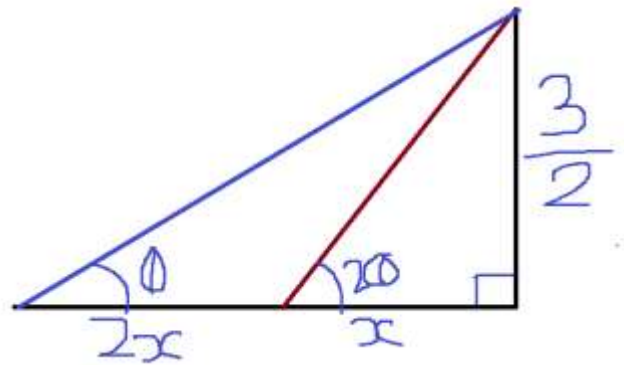
$$\tan \theta = \frac{\frac{3}{2}}{3x} \Rightarrow x = \frac{1}{2 \tan \theta}$$

From the tent and the pole:

$$\tan 2\theta = \frac{\frac{3}{2}}{x} \Rightarrow x = \frac{3}{2 \tan 2\theta}$$

But both values of x must be equal. Hence:

$$\frac{1}{2 \tan \theta} = \frac{3}{2 \tan 2\theta} \Rightarrow \tan 2\theta = 3 \tan \theta$$



Use the double angle identity for $\tan \theta$, and then solve for θ :

$$\begin{aligned} \frac{2 \tan \theta}{1 - \tan^2 \theta} &= 3 \tan \theta \\ 2 &= 3 - 3 \tan^2 \theta \\ \tan^2 \theta &= \frac{1}{3} \\ \tan \theta &= \frac{1}{\sqrt{3}} \\ \theta &= 30^\circ \\ \sin \theta &= \frac{1}{2} \end{aligned}$$

Use the formula for $\sin \theta$ in the triangle with the rope:

$$\sin \theta = \frac{\text{opp}}{\text{hyp}} = \frac{l(\text{pole})}{l(\text{rope})} = \frac{\frac{3}{2}}{l(\text{rope})} = \frac{1}{2} \Rightarrow l(\text{rope}) = 3$$

2.17: Sum and Difference Identities

$$\sin(\alpha \pm \beta) = \sin \alpha \cos \beta \pm \cos \alpha \sin \beta$$

$$\cos(\alpha \pm \beta) = \cos \alpha \cos \beta \mp \sin \alpha \sin \beta$$

Note that

- Sine of a sum, RHS has a plus sign.
- Sine of a difference, RHS has a negative sign
- Cosine of a sum, RHS has a plus sign.
- Cosine of a difference, RHS has a negative sign

Example 2.18

Mark the correct option

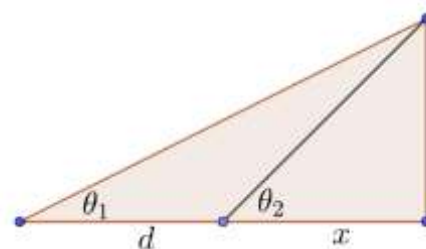
The angle of elevation of the top of a vertical tower of a point P on the horizontal ground was observed to be α . After moving a distance 2 meters from P towards the foot of the tower, the angle of elevation changes to β . Then the height (in meters) of the tower, is:

- A. $\frac{2 \sin \alpha \sin \beta}{\sin(\beta - \alpha)}$
- B. $\frac{2 \sin \beta}{\cos(\beta - \alpha)}$
- C. $\frac{2 \sin(\beta - \alpha)}{\sin \alpha \sin \beta}$
- D. $\frac{\cos(\beta - \alpha)}{\sin \alpha \sin \beta}$ (JEE Main, April 11, 2014)

$$h = d \frac{\tan \theta_2 \tan \theta_1}{\tan \theta_2 - \tan \theta_1}, \theta_2 > \theta_1$$

Substitute $\theta_2 = \beta, \theta_1 = \alpha, d = 2$ in the above. Expand using $\tan \theta = \frac{\sin \theta}{\cos \theta}$:

$$= 2 \frac{\frac{\sin \beta}{\cos \beta} \cdot \frac{\sin \alpha}{\cos \alpha}}{\frac{\sin \beta}{\cos \beta} - \frac{\sin \alpha}{\cos \alpha}}$$



Add the fractions and cancel the denominator:

$$= 2 \frac{\frac{\sin \beta \sin \alpha}{\cos \beta \cos \alpha}}{\frac{\sin \beta \cos \alpha - \sin \alpha \cos \beta}{\cos \beta \cos \alpha}} = 2 \frac{\sin \beta \sin \alpha}{\sin \beta \cos \alpha - \sin \alpha \cos \beta}$$

Use the property $\sin(\alpha - \beta) = \sin \alpha \cos \beta - \cos \alpha \sin \beta$. The positions of α and β are interchanged. Hence:

$$= 2 \frac{\sin \beta \sin \alpha}{\sin(\beta - \alpha)} \Rightarrow \text{Option A}$$

D. Time, Speed and Distance

Example 2.19

A man is walking towards a vertical pillar in a straight path, at a uniform speed. At a certain point A on the path, he observes that the angle of elevation of the top of the pillar is 30° . After walking for 10 minutes from A in the same direction, at a point B , he observes that the angle of elevation of the top of the pillar is 60° . Then the time

taken (in minutes) by him, from B to reach the pillar is: (JEE Main 2016)

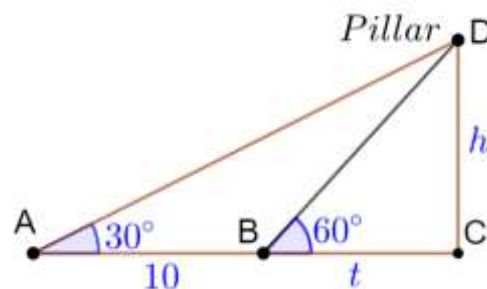
Distance is proportional to time, so let

$B = 10 \text{ meters}, BC = t \text{ meters} \Rightarrow \text{Will take } t \text{ minutes}$

$$\text{In } \triangle DAC: \tan 30^\circ = \frac{1}{\sqrt{3}} = \frac{h}{10+t} \Rightarrow h = \frac{10+t}{\sqrt{3}}$$

$$\text{In } \triangle DBC: \tan 60^\circ = \sqrt{3} = \frac{h}{t} \Rightarrow h = \sqrt{3}t$$

$$\frac{10+t}{\sqrt{3}} = \sqrt{3}t \Rightarrow t = 5$$



Example 2.20

A bird is sitting on top of a vertical pole 20 m which makes an angle of elevation 45° from a point O on the ground. It flies off horizontally straight away from the point O . After one second, the elevation of the bird from O is reduced to 30° . Then the speed (in $\frac{m}{s}$) of the bird is: (JEE Main 2014)

Since we have a $45-45-90$ triangle

Distance from O to Pole = 20 m

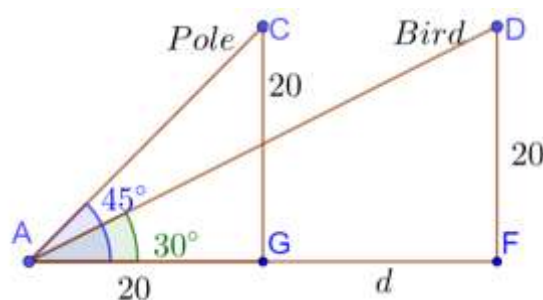
In $\triangle ADF$

$$\tan 30^\circ = \frac{1}{\sqrt{3}} = \frac{20}{20+d}$$

$$20+d = 20\sqrt{3}$$

$$d = 20(\sqrt{3}-1)$$

$$\text{Speed} = \frac{\text{Distance}}{\text{Time}} = \frac{20(\sqrt{3}-1)}{1} = 20(\sqrt{3}-1) \frac{m}{s}$$



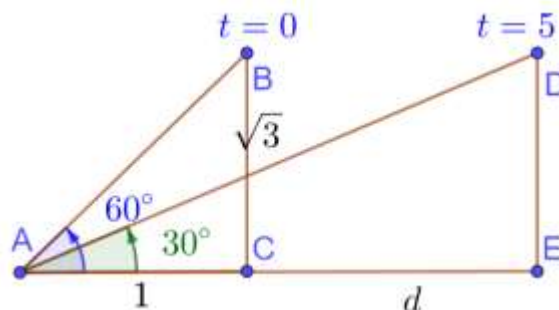
Example 2.21

An aeroplane flying at a constant speed, parallel to the horizontal ground, $\sqrt{3} \text{ km}$ above it is observed at an elevation of 60° from a point on the ground. If, after five seconds, its elevation at the same point, is 30° , then the speed (in $\frac{km}{hr}$) of the aeroplane is: (JEE-M 2018)

$$\triangle ABC: \tan 60^\circ = \frac{BC}{AC} \Rightarrow AC = \frac{BC}{\tan 60^\circ} = \frac{\sqrt{3}}{\sqrt{3}} = 1$$

$$\triangle ADE: \tan 30^\circ = \frac{\sqrt{3}}{1+d} \Rightarrow \frac{1}{\sqrt{3}} = \frac{\sqrt{3}}{1+d} = 1+d = 3 \Rightarrow d = 2$$

$$S = \frac{d}{t} = \frac{2 \text{ km}}{5 \text{ s}} = \frac{2}{5} \times 3600 \frac{\text{km}}{\text{hr}} = 1440 \frac{\text{km}}{\text{hr}}$$



Example 2.22

A bomber flying h feet high has an angle of depression of θ to its target, and releases a missile flying at $s_1 \frac{m}{s}$. The plane is exactly above a military station, from which troops leave to reach the target at the same time as the missile is released. The troops have a speed of $s_2 \frac{m}{s}$. The missile reaches the target first. The troops will reach $t = h \left(\frac{f_1(\theta)}{s_2} - \frac{f_2(\theta)}{s_1} \right)$ seconds later. Find the value of t .

Angle of Elevation = Angle of Depression = θ

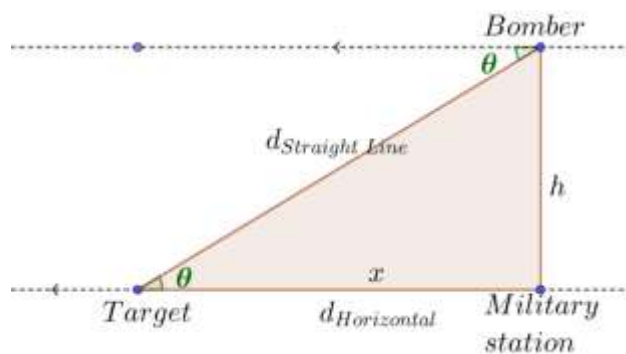
We use the trig ratios to find the values that we want¹

$$\cot \theta = \frac{d_{\text{horizontal}}}{h} \Rightarrow d_{\text{horizontal}} = h \cot \theta$$

$$\sin \theta = \frac{h}{d_{SL}} \Rightarrow d_{SL} = h \csc \theta$$

Using $T = \frac{D}{S}$, the time difference

$$= \underbrace{\frac{d_{\text{horizontal}}}{S_2}}_{\text{Troop Time}} - \underbrace{\frac{d_{SL}}{S_1}}_{\text{Missile Time}} = \frac{h \cot \theta}{S_2} - \frac{h \csc \theta}{S_1} = h \left(\frac{\cot \theta}{S_2} - \frac{\csc \theta}{S_1} \right)$$



E. Two Triangles: Vertical Movement

Pending

Example 2.23

43: From the base of a pole of height 20 meter, the angle of elevation of the top of a tower is 60° . The pole subtends an angle of 30° at the top of the tower. Then the height of the tower is: (JEE-M 2022)

2.24: Trig Functions of 15° and 75°

$$\sin 15^\circ = \cos 75^\circ = \frac{\sqrt{3} - 1}{2\sqrt{2}}$$

$$\cos 15^\circ = \sin 75^\circ = \frac{\sqrt{3} + 1}{2\sqrt{2}}$$

$$\tan 15^\circ = 2 - \sqrt{3}$$

$$\tan 75^\circ = 2 + \sqrt{3}$$

Example 2.25

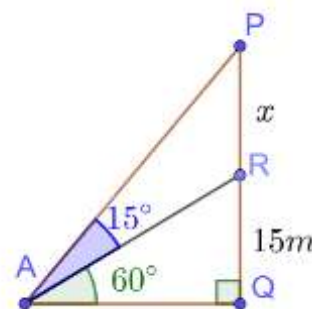
A tower PQ stands on a horizontal ground with base Q on the ground. The point R divides the tower into parts such that $QR = 15m$. If from a point A on the ground the angle of elevation of R is 60° , and the part PR of the tower subtends an angle of 15° at A, then the height of the tower is: (JEE-M 2022)

$$\tan 60^\circ = \frac{15}{AQ} \Rightarrow AQ = \frac{15}{\tan 60^\circ} = \frac{15}{\sqrt{3}} = \frac{15\sqrt{3}}{3} = 5\sqrt{3}$$

$$\tan 75^\circ = \frac{PQ}{AQ}$$

$$\frac{PQ}{5\sqrt{3}} = 2 + \sqrt{3}$$

$$PQ = 10\sqrt{3} + 15$$



¹ If you use $\tan \theta = \frac{d_{\text{horizontal}}}{h}$, then your trigonometric function will be in the denominator, and you will have to manipulate to move it to the numerator.

2.26: Double Angle Identity for $\tan$

$$\tan 2\theta = \frac{2 \tan \theta}{1 - \tan^2 \theta}$$

Example 2.27

A vertical pole fixed to the horizontal ground is divided in the ratio 3: 7 by a mark on it with lower part shorter than the upper part. If the two parts subtend equal angles at a point on the ground 18m away from the base of the pole, then the height of the pole is (as a number): (JEE-M 2017, 2021)

In $\triangle ABC$

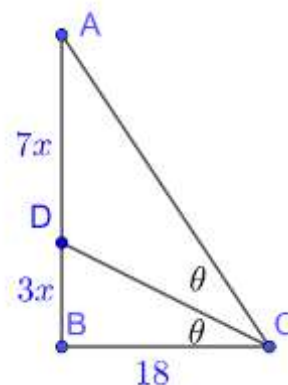
$$\tan 2\theta = \frac{2 \tan \theta}{1 - \tan^2 \theta} = \frac{10x}{18}$$

Substitute $\tan \theta = \frac{3x}{18} = \frac{x}{6}$ from $\triangle DBC$:

$$\frac{2 \left(\frac{x}{6} \right)}{1 - \left(\frac{x}{6} \right)^2} = \frac{10x}{18}$$

$$6x = 10x - \frac{10x^3}{36}$$

$$\frac{10x^3}{36} = 4x \Rightarrow 10x^2 = 144 \Rightarrow \sqrt{10}x = 12 \Rightarrow x = 12\sqrt{10}$$



2.28: Sum and Difference Identities for $\tan$

$$\tan(\alpha + \beta) = \frac{\tan \alpha + \tan \beta}{1 - \tan \alpha \tan \beta}$$

$$\tan(\alpha - \beta) = \frac{\tan \alpha - \tan \beta}{1 + \tan \alpha \tan \beta}$$

Example 2.29

The upper $\frac{3}{4}$ portion of a vertical pole subtends an angle $\tan^{-1} \frac{3}{5}$ at a point in the horizontal plane through its foot and at a distance of 40m from the foot. The possible height(s) of the vertical pole are (as a number): (JEE-M 2003, Adapted)

$$\alpha + \beta = \theta \Rightarrow \beta = \theta - \alpha \Rightarrow \tan \beta = \tan(\theta - \alpha)$$

RHS

Substitute $\beta = \tan^{-1} \frac{3}{5}$

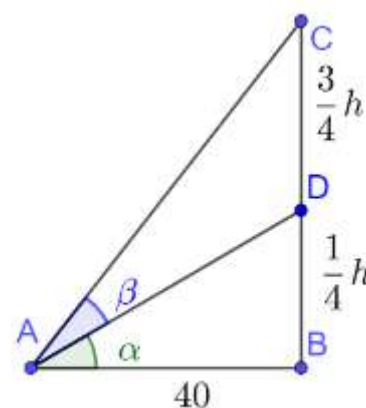
$$\tan \beta = \tan \left(\tan^{-1} \frac{3}{5} \right) = \frac{3}{5}$$

LHS

Use the difference of angles formula:

$$\tan(\theta - \alpha) = \frac{\tan \theta - \tan \alpha}{1 + \tan \theta \tan \alpha}$$

Substitute $\tan \theta = \frac{h}{40}$, $\tan \alpha = \frac{\frac{1}{4}h}{40} = \frac{h}{160}$:



$$= \frac{\frac{h}{40} - \frac{h}{160}}{1 + \left(\frac{h}{40}\right)\left(\frac{h}{160}\right)} = \frac{\frac{3h}{160}}{1 + \frac{h^2}{6400}} = \frac{3h}{160} \cdot \frac{6400}{6400 + h^2} = \frac{3(40h)}{6400 + h^2}$$

Combine LHS and RHS, cross-multiply and solve the resulting quadratic:

$$\frac{3}{5} = \frac{3(40h)}{6400 + h^2} \Rightarrow h^2 - 200h + 6400 = 0 \Rightarrow h \in \{40, 160\}$$

Example 2.30

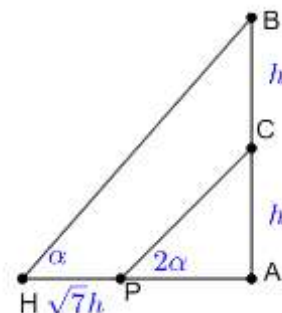
Let a vertical tower AB of height $2h$ stands on a horizontal ground. Let from a point P on the ground a man can see upto height h of the tower with an angle of elevation 2α . When from P , he moves a distance d in the direction of AP he can see the top B of the tower with an angle of elevation α . If $d = \sqrt{7}h$, then the possible values of $\tan \alpha$ are: (answer as a number): (JEE-Main 2022, Adapted)

In $\triangle CPA$

$$\tan 2\alpha = \frac{h}{PA} \Rightarrow PA = \frac{h}{\tan 2\alpha} = \frac{h}{\frac{2 \tan \alpha}{1 - \tan^2 \alpha}} = \frac{h(1 - \tan^2 \alpha)}{2 \tan \alpha}$$

In $\triangle BHA$:

$$\tan \alpha = \frac{2h}{PA + \sqrt{7}h} \Rightarrow PA = \frac{2h}{\tan \alpha} - \sqrt{7}h = \frac{h(2 - \sqrt{7}(\tan \alpha))}{\tan \alpha}$$



Equate the RHS of Equations I and II and substitute $x = \tan \alpha$:

$$\begin{aligned} \frac{h(1 - x^2)}{2x} &= \frac{2h - \sqrt{7}hx}{x} \\ 1 - x^2 &= 4 - 2\sqrt{7}x \\ x^2 - 2\sqrt{7}x + 3 &= 0 \\ \tan \alpha = x &= \frac{2\sqrt{7} \pm \sqrt{28 - (4)(1)(3)}}{2} = \frac{2\sqrt{7} \pm 4}{2} = \sqrt{7} \pm 2 \end{aligned}$$

Example 2.31

From the top A of a vertical wall AB of height $30m$, the angles of depression of the top P and bottom Q of a vertical tower PQ are 15° and 60° respectively. B and Q are on the same horizontal level. If C is a point on AB such that $CB = PQ$, then the area (in m^2) of the quadrilateral $BCPQ$ is (answer as a number): (JEE Main, April 6, 2023, Shift-I)

In $\triangle ABQ$

$$\tan 60^\circ = \sqrt{3} = \frac{30}{BQ} \Rightarrow BQ = 10\sqrt{3}$$

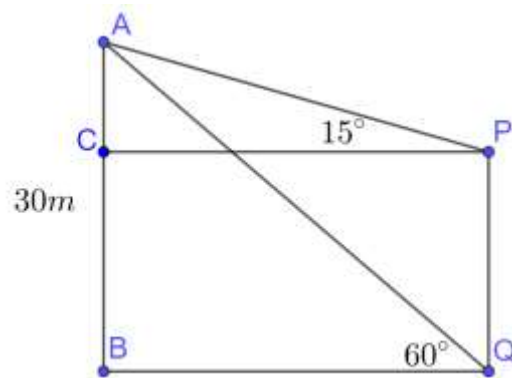
In $\triangle ACP$

$$\tan 15^\circ = 2 - \sqrt{3} = \frac{AC}{CP}$$

$$AC = (2 - \sqrt{3})CP = (2 - \sqrt{3})(10\sqrt{3}) = 20\sqrt{3} - 30$$

$$CB = 30 - AC = 30 - (20\sqrt{3} - 30) = 60 - 20\sqrt{3}$$

$$(BQ)(CB) = 10\sqrt{3}(60 - 20\sqrt{3}) = 600\sqrt{3} - 600$$



2.2 Heights and Distances: II

A. Optics

Example 2.32

50: A ray of light coming from the point $(2, 2\sqrt{3})$ is incident at an angle 30° on the line $x = 1$ at the point A. The ray gets reflected on the line $x = 1$ and meet the x -axis at the point B. Then, the line AB passes through the point:

- A. $(3, -\frac{1}{\sqrt{3}})$
- B. $(4, -\frac{\sqrt{3}}{2})$
- C. $(3, -\sqrt{3})$
- D. $(4, -\sqrt{3})$ (JEE Main Sep 6, 2022, Shift-I)

$$\text{Slope of } AB = \frac{y_2 - y_1}{x_2 - x_1} = -\sqrt{3}$$

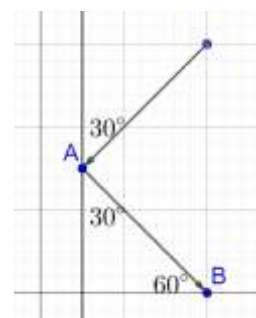
$$y - y_1 = m(x - x_1)$$

$$y - 0 = -\sqrt{3}(x - 2)$$

$$y = -\sqrt{3}x + 2\sqrt{3}$$

Check the options and substitute

Option C



B. Looking Up and Down

Example 2.33

A house of height 100 m subtends a right angle at the window of an opposite house. If the height of the window be 64 m , then the distance between the two houses is: (NTA Abhyas, JEE/Math/Heights and Distances)

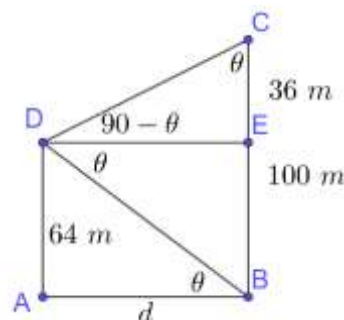
Let d be the distance between the two houses.

$$\text{In right } \triangle DAB: \tan \theta = \frac{64}{d}$$

$$\text{In right } \triangle CDE \tan \theta = \frac{d}{36}$$

Equate the two values:

$$\begin{aligned} \frac{64}{d} &= \frac{d}{36} \\ d^2 &= 64 \times 36 \\ d &= 8 \times 6 = 48 \end{aligned}$$



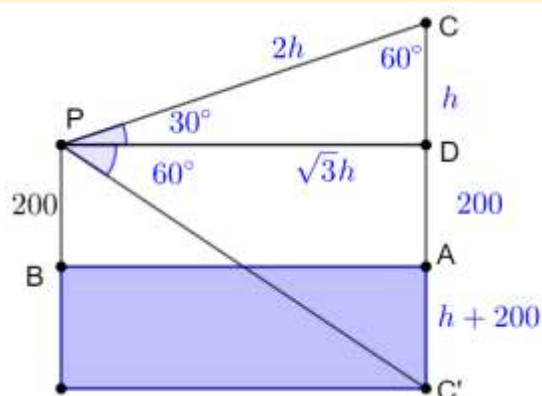
Example 2.34

The angle of elevation of a cloud C from a point P , 200 m above a still lake is 30° . If the angle of depression of the image of C in the lake from the point P is 60° , then PC (in meters) is equal to: (JEE-M 2020)

In $\triangle PDC'$:

$$\tan 60^\circ = \sqrt{3} = \frac{h + 400}{\sqrt{3}h} \Rightarrow 3h = h + 400 \Rightarrow 2h = 400$$

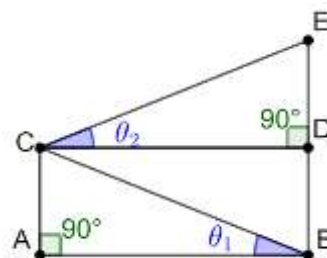
$$PC = 400$$



Example 2.35

In the figure, $\theta_1 + \theta_2 = \frac{\pi}{2}$ and $\sqrt{3}(BE) = 4(AB)$. If the area of $\triangle CAB$ is $2\sqrt{3} - 3 \text{ unit}^2$, when $\frac{\theta_2}{\theta_1}$ is the largest, then the perimeter (in units) of $\triangle CED$ is equal to:

(JEE-M 2023)



The given condition has a relation between BE and AB . Hence, write the parts of BE in terms of AB :

$$\tan \theta_1 = \frac{AC}{AB} \Rightarrow BD = AC = AB \tan \theta_1$$

Equation I

$$\angle CED = \theta_1 \Rightarrow \tan \theta_1 = \frac{CD}{DE} \Rightarrow DE = \frac{CD}{\tan \theta_1} = \frac{AB}{\tan \theta_1}$$

Equation II

$$BE = BD + DE = AB \tan \theta_1 + \frac{AB}{\tan \theta_1} = AB \left(\tan \theta_1 + \frac{1}{\tan \theta_1} \right)$$

Substitute $\sqrt{3}(BE) = 4(AB) \Rightarrow BE = \frac{4}{\sqrt{3}}AB$:

$$\begin{aligned} \frac{4}{\sqrt{3}}AB &= AB \left(\tan \theta_1 + \frac{1}{\tan \theta_1} \right) \\ \frac{4}{\sqrt{3}} &= \frac{3 + 1}{\sqrt{3}} = \sqrt{3} + \frac{1}{\sqrt{3}} = \tan \theta_1 + \frac{1}{\tan \theta_1} \end{aligned}$$

Hence, the solutions are²:

$$\tan \theta_1 = x \in \left\{ \sqrt{3}, \frac{1}{\sqrt{3}} \right\}$$

To make $\frac{\theta_2}{\theta_1}$ largest, we make $\tan \theta_1$ smallest, giving $\tan \theta_1 = \frac{1}{\sqrt{3}} \Rightarrow \theta_1 = 30^\circ \Rightarrow \theta_2 = 60^\circ$.

Use the area of the triangle condition:

$$2\sqrt{3} - 3 = [CAB] = \frac{1}{2}(AB)(AC) = \frac{1}{2}(AB)(AB \tan \theta_1) = \frac{1}{2}AB^2 \left(\frac{1}{\sqrt{3}} \right)$$

$$AB^2 = 2\sqrt{3}(2\sqrt{3} - 3) = 12 - 6\sqrt{3}$$

Use $(a - b)^2 = a^2 - 2ab + b^2$

$$2ab = 6\sqrt{3} \Rightarrow ab = 3\sqrt{3} \Rightarrow CD = AB = 3 - \sqrt{3}$$

Using the properties of a 30 - 60 - 90 triangle:

$$\begin{aligned} \text{Perimeter} &= CD + DE + CE = CD + \sqrt{3}CD + 2CD = CD(1 + \sqrt{3} + 2) = CD(3 + \sqrt{3}) \\ &= (3 - \sqrt{3})(3 + \sqrt{3}) = 9 - 3 = 6 \end{aligned}$$

C. Two Poles

2.36: Trig Functions of 22.5 and 67.5

$$\begin{aligned} \sin 22.5^\circ &= \cos 67.5^\circ = \frac{\sqrt{2} - 1}{\sqrt{4 - 2\sqrt{2}}} \\ \cos 22.5^\circ &= \sin 67.5^\circ = \frac{1}{\sqrt{4 - 2\sqrt{2}}} \\ \tan 22.5^\circ &= \sqrt{2} - 1 \end{aligned}$$

Example 2.37

44: Let AB and PQ be two vertical poles, 160m apart from each other. Let C be the middle point of B and Q , which are feet of the two poles. Let $\frac{\pi}{8}$ and θ be the angles of elevation from C to P and A , respectively. If the height of Pole PQ is twice the height of pole AB , then $\tan^2 \theta$ is equal to: (JEE-M 2022)

² We can also do this by solving the equation. Substitute $\tan \theta_1 = x$ to get $\frac{4}{\sqrt{3}} = x + \frac{1}{x} \Rightarrow \sqrt{3}x^2 - 4x + \sqrt{3} = 0$

Factor using $Product = \sqrt{3}\sqrt{3} = 3, Sum = -4 = -3 - 1$:

$$\begin{aligned} \sqrt{3}x^2 - 3x - x + \sqrt{3} &= 0 \\ \sqrt{3}x(x - \sqrt{3}) - 1(x - \sqrt{3}) &= 0 \\ (\sqrt{3}x - 1)(x - \sqrt{3}) &= 0 \end{aligned}$$

In $\triangle ABC$

$$\tan \theta = \frac{h}{80} \Rightarrow h = 80 \tan \theta$$

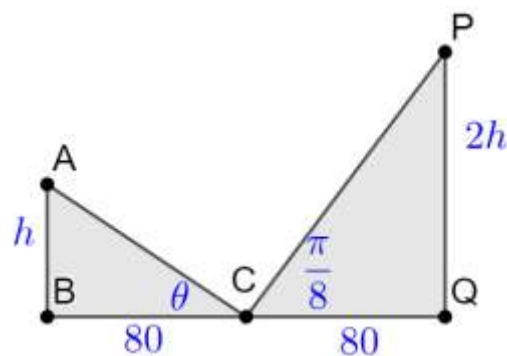
In $\triangle PQC$

$$\tan \frac{\pi}{8} = \frac{2h}{80} \Rightarrow h = 40 \tan \frac{\pi}{8}$$

$$80 \tan \theta = 40 \tan \frac{\pi}{8}$$

$$\tan \theta = \frac{1}{2} \tan \frac{\pi}{8} = \frac{1}{2} (\sqrt{2} - 1)$$

$$\tan^2 \theta = \frac{1}{4} (2 - 2\sqrt{2} + 1) = \frac{1}{4} (3 - 2\sqrt{2})$$



Pending

Example 2.38

48: Two vertical poles are 150m apart and the height of one is three times that of the other. If from the middle point of the line joining their feet, an observer finds the angles of elevation of their tops to be complementary, then the height of the shorter pole (in meters) is: (JEE-M 2021)

2.39: Similarity Property

In $\triangle DAC$ and $\triangle PAF$

$$\begin{aligned} \angle DCA &= \angle PFA = 90^\circ \\ \angle DAC &= \angle PAF \text{ (Same Angle)} \\ \triangle DAC &\sim \triangle PAF \end{aligned}$$

$$\frac{m}{m+n} = \frac{PF}{CD} \Rightarrow PF = CD \left(\frac{m}{m+n} \right) \quad \text{Equation I}$$

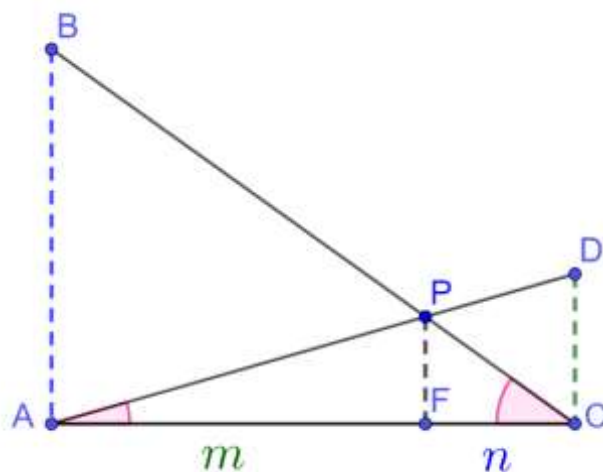
In $\triangle BAC$ and $\triangle PFC$

$$\begin{aligned} \angle BAC &= \angle PFC = 90^\circ \\ \angle BCA &= \angle PCF \text{ (Same Angle)} \\ \triangle BAC &\sim \triangle PCF \end{aligned}$$

$$\frac{n}{m+n} = \frac{PF}{AB} \Rightarrow PF = AB \left(\frac{n}{m+n} \right) \quad \text{Equation II}$$

From Equation I and II:

$$\begin{aligned} CD \left(\frac{m}{m+n} \right) &= AB \left(\frac{n}{m+n} \right) \\ \frac{m}{n} &= \frac{AB}{CD} \end{aligned}$$

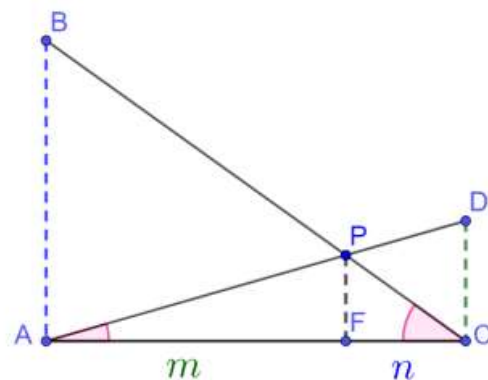


Example 2.40

52: Two vertical poles $AB = 15m$ and $CD = 10m$ are standing apart on a horizontal ground with points A and C on the ground. If P is the point of intersection of BC and AD , then the height of P (in meters) above the line AC is: (JEE-M 2020)

$$\frac{m}{n} = \frac{AB}{CD} = \frac{15}{10} = \frac{3}{2} \Rightarrow m = \frac{3}{2}n$$

$$PF = CD \left(\frac{m}{m+n} \right) = 10 \left(\frac{\frac{3}{2}n}{\frac{3}{2}n + n} \right) = 10 \left(\frac{\frac{3}{2}n}{\frac{5}{2}n} \right) = 10 \left(\frac{3}{5} \right) = 6$$



D. Three Triangles: Multiple Angles of Elevation

Two triangles increase the complexity of a question. Three triangles increase the complexity of a question still further.

Example 2.41

If the angles of elevation of the top of a tower from three collinear points A, B and C, on a line leading to the foot of the tower are 30° , 45° and 60° , respectively, then the ratio AB : BC is: (JEE-M 2015)

Method I

Without loss of generality, let the height of the tower be 1. Then

$$QA = \frac{1}{\tan 30^\circ} = \sqrt{3}, QB = \frac{1}{\tan 45^\circ} = 1, QC = \frac{1}{\tan 60^\circ} = \frac{1}{\sqrt{3}}$$

$$\frac{AB}{BC} = \frac{QA - QB}{QB - QC} = \frac{\sqrt{3} - 1}{1 - \frac{1}{\sqrt{3}}} = \frac{\sqrt{3} - 1}{\frac{\sqrt{3} - 1}{\sqrt{3}}} = \sqrt{3}$$

Method II: Longer

Let $QC = a$, $CB = b$, $AB = c \Rightarrow \frac{AB}{BC} = \frac{c}{b}$

$$\tan 60^\circ = \frac{h}{a} \Rightarrow h = a \tan 60^\circ = \sqrt{3}a$$

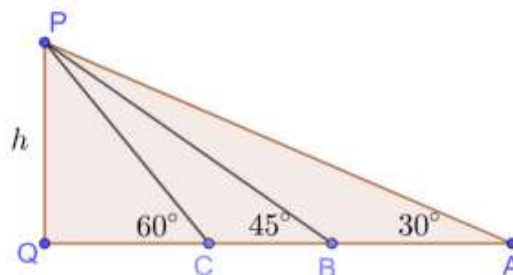
Equation I

$$\tan 45^\circ = \frac{h}{a+b} \Rightarrow h = (a+b) \tan 45^\circ = a+b$$

Equation II

$$\tan 30^\circ = \frac{h}{a+b+c} \Rightarrow h = (a+b+c) \tan 30^\circ = \frac{a+b+c}{\sqrt{3}}$$

Equation III



From Equations I and II:

$$\sqrt{3}a = a+b \Rightarrow (\sqrt{3}-1)a = b \Rightarrow a = \frac{b}{\sqrt{3}-1} = \frac{b}{2}(\sqrt{3}+1)$$

Equation IV

From Equations I and III:

$$\sqrt{3}a = \frac{a+b+c}{\sqrt{3}} \Rightarrow 3a = a+b+c \Rightarrow a = \frac{b+c}{2}$$

Equation V

From Equations IV and V

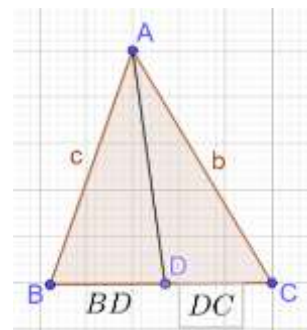
$$\frac{b}{2}(\sqrt{3} + 1) = \frac{b + c}{2} \Rightarrow b(\sqrt{3} + 1) = b + c \Rightarrow \sqrt{3}b = c \Rightarrow \frac{c}{b} = \sqrt{3}$$

2.42: Angle Bisector Theorem

An angle bisector of a triangle divides the side opposite it in the ratio of the other two sides of the triangle.

In the diagram, if AD is a bisector of $\angle BAC$:

$$\frac{BD}{c} = \frac{DC}{b} \Leftrightarrow \frac{BD}{DC} = \frac{c}{b} \Leftrightarrow BD:DC = c:b$$



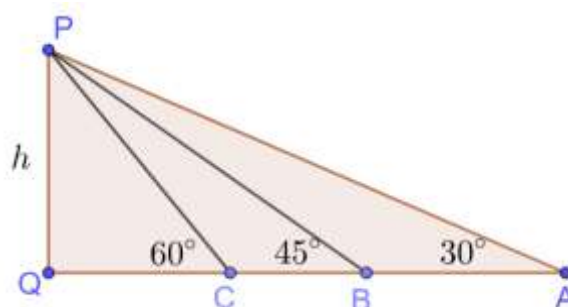
Example 2.43

If the angles of elevation of the top of a tower from three collinear points A , B and C , on a line leading to the foot of the tower are 30° , 45° and 60° , respectively, then the ratio $AB:BC$ is: (JEE-M 2015)

$$\begin{aligned}\angle PCB &= 120^\circ \\ \angle CPB &= 180 - 120 - 45 = 15^\circ \\ \angle BPA &= 180 - 135 - 30 = 15^\circ \\ \angle CPB &= \angle BPA = 15^\circ \Rightarrow PB \text{ bisects } \angle CPA\end{aligned}$$

By the angle bisector theorem:

$$\begin{aligned}AB:BC &= PA:PC \\ \sin 30^\circ &= \frac{h}{PA} \Rightarrow h = \frac{PA}{2} \\ \sin 60^\circ &= \frac{h}{PC} \Rightarrow h = \frac{\sqrt{3}}{2}PC \\ \frac{PA}{2} &= \frac{\sqrt{3}}{2}PC \Rightarrow \frac{PA}{PC} = \sqrt{3} \Rightarrow PA:PC = \sqrt{3}:1\end{aligned}$$

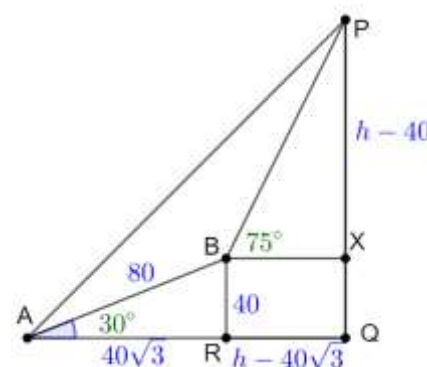


E. Three Triangles: Incline

Example 2.44

The angle of elevation of the top of hill from a point on the horizontal plane passing through the foot of the hill is found to be 45° . After walking a distance of 80 meters towards the slope, up a slope inclined at an angle of 30° to the horizontal plane, the angle of elevation of the top of the hill becomes 75° . Then the height of the hill (in meters) is (JEE-M 2020)

$$\begin{aligned}\tan 75^\circ &= 2 + \sqrt{3} = \frac{h - 40}{h - 40\sqrt{3}} \\ 2h - 80\sqrt{3} + \sqrt{3}h - 120 &= h - 40 \\ 2h - 80\sqrt{3} + \sqrt{3}h - 120 &= h - 40 \\ h + \sqrt{3}h &= 80 + 80\sqrt{3} \\ h(1 + \sqrt{3}) &= 80(1 + \sqrt{3}) \\ h &= 80\end{aligned}$$



Example 2.45

The angle of elevation of the top of P of a vertical tower PQ of height 10 from a point A on the horizontal ground is 45° . Let R be a point on AQ and from a point B , vertically above R , the angle of elevation is 60° . If $\angle BAQ =$

30° , $AB = d$ and the area of the trapezium $PQRB$ is α , then the ordered pair (d, α) is (Answer as a numerical expression) (JEE-M 2022)

Since $\triangle ABR$ is a $30 - 60 - 90$ triangle, substitute $AR = \frac{\sqrt{3}}{2}d$, $BR = XQ = \frac{d}{2}$:

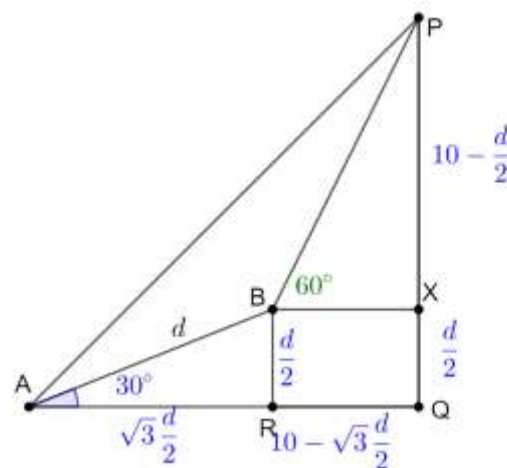
$$\tan 60^\circ = \frac{PX}{BX} = \frac{PQ - XQ}{QA - RQ} = \frac{10 - \frac{d}{2}}{10 - \frac{\sqrt{3}}{2}d} = \frac{\frac{20 - d}{2}}{\frac{20 - \sqrt{3}d}{2}} = \frac{20 - d}{20 - \sqrt{3}d}$$

Substitute $\tan 60^\circ = \sqrt{3}$:

$$\sqrt{3} = \frac{20 - d}{20 - \sqrt{3}d} \Rightarrow 20\sqrt{3} - 3d = 20 - d \Rightarrow d = 10\sqrt{3} - 10$$

The area of the trapezium $PQRB$

$$\begin{aligned} &= (RQ) \left(\frac{BR + QP}{2} \right) = \left(\frac{20 - \sqrt{3}d}{2} \right) \left(\frac{d + 20}{4} \right) \\ &= \left(\frac{20 - \sqrt{3}(10\sqrt{3} - 10)}{2} \right) \left(\frac{(10\sqrt{3} - 10) + 20}{4} \right) \\ &= \frac{1}{8} (10\sqrt{3} - 10)(10\sqrt{3} + 10) = \frac{1}{8} (300 - 100) = \frac{200}{8} = 25 \end{aligned}$$



F. Tangents to a Circle

2.46: Tangents to a Circle

- The tangents to a circle drawn from a point outside the circle are congruent.
- Tangent Perpendicularity Theorem: The tangent to a circle is perpendicular to the radius at the point of tangency.

Example 2.47

A spherical balloon of radius r subtends an angle α at the eye of an observer. If the angle of elevation of the center of the balloon be β , then the height of the center of the balloon in terms of r, α, β is: (JEE-M 2021, Adapted)

Let

O = center of balloon

l = Distance from Observer to Balloon

h = height of balloon

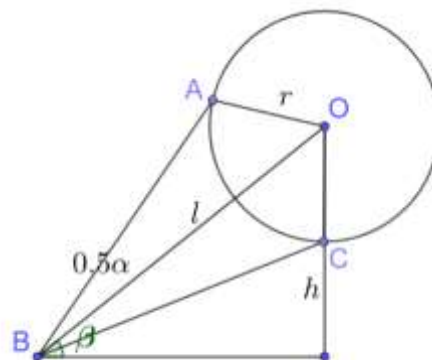
$\alpha = \angle ABC$

The lines drawn from point B to the sphere are tangent to the sphere. The radii at the point of contact are perpendicular to the tangents (by the tangent perpendicularity theorem).

$$\triangle BAO \cong \triangle BCO \Rightarrow \angle ABO = \angle CBO = \frac{\alpha}{2} = \frac{\alpha}{2}$$

$$\text{In right } \triangle ABO: \sin\left(\frac{\alpha}{2}\right) = \frac{r}{l} \Rightarrow l = \frac{r}{\sin\left(\frac{\alpha}{2}\right)}$$

$$\text{In right } \triangle OBD: \sin \beta = \frac{h}{l} \Rightarrow h = l \sin \beta = \frac{r \sin \beta}{\sin\left(\frac{\alpha}{2}\right)}$$



2.48: Absolute and Percentage Error

Absolute error is the difference between an actual value and an approximated value:

$$\text{Absolute Error} = |r - r_{\text{Approx}}|$$

Percentage is absolute error written as a percentage

$$\text{Percentage Error} = \left| \frac{r - r_{\text{Approx}}}{r} \right| \times 100$$

Example 2.49: Radius of the Earth

- Show that if the angle of depression from a mountain with height h meters to the ground is α° , then the radius of earth is $r = \frac{h \cos \alpha}{1 - \cos \alpha}$.
- Estimate the radius of the earth in km if $h = 965$ and $\alpha = 89^\circ$.
- Use the actual radius of the earth (6371 km) to find the absolute error and the percentage error in the estimation in Part B.

Draw the Earth, with a mountain on top. Let:

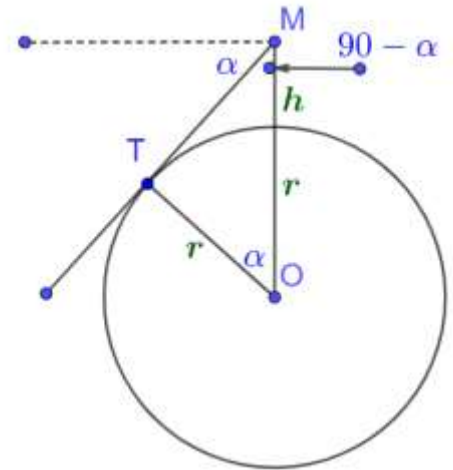
O = Center of Earth
 M = Tip of Mountain
 r = Radius of Earth
 h = height of the mountain
 α = angle of depression

Note that from the mountain you can see upto T, and hence:

MT is tangent to the Earth

By the tangent perpendicularity theorem:

$$\begin{aligned}\angle MTO &= 90^\circ \\ \angle TMO &= 90 - \alpha \\ \angle TOM &= 180 - 90 - (90 - \alpha) = \alpha\end{aligned}$$



Part A

In right $\triangle MTO$:

$$\cos \alpha = \frac{r}{r + h}$$

Clear fractions:

$$r \cos \alpha + h \cos \alpha = r$$

Collate all r terms on the RHS, and factor out r

$$h \cos \alpha = r(1 - \cos \alpha)$$

Solve for r :

$$r = \frac{h \cos \alpha}{1 - \cos \alpha}$$

Part B

$$r = \frac{h \cos \alpha}{1 - \cos \alpha} = \frac{965 \cos 1^\circ}{1 - \cos 1^\circ} = 6335 \text{ km}$$

Part C

$$\text{Absolute Error} = |6371 - 6335| = |36| = 36$$

$$\text{Percentage Error} = \left| \frac{r - r_{\text{Approx}}}{r} \right| \times 100 = \frac{36}{6371} \times 100 = 0.56\%$$

Example 2.50

Measuring Diameter of a Pipe

G. Multiple Triangles

Example 2.51

2.3 3D Problem Solving

A. Cross Sections

A standard approach to 3D problems is take one or more suitable 2D cross sections, work with them, and arrive at the required answer. This technique is powerful because it reduces a difficult problem into a known problem.

2.52: Cross Section

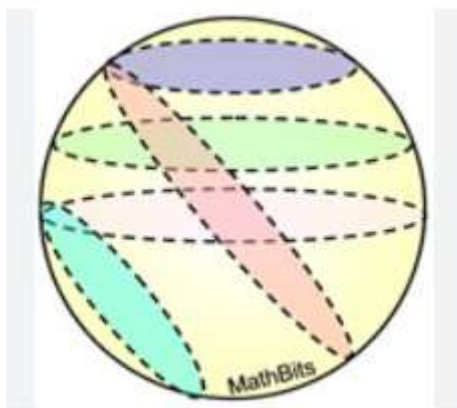
A cross section is a 2D “slice” of a 3D object

When working with three dimensional objects, drawing is a challenge. One way to simplify is draw a 2D cross section of a 3D object.

Example 2.53

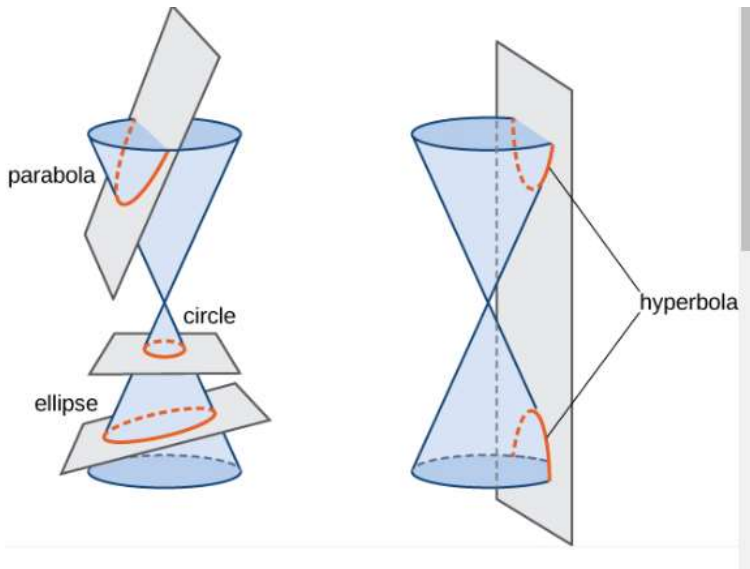
Draw a cross section of a sphere.

The cross section of a sphere is always a circle.



Example 2.54

The conics (parabola, hyperbola, circle and ellipse) are cross section of two right circular cones placed with tips together.



Example 2.55

What is the cross section of a right square pyramid:

- A. Parallel to the Base
- B. Perpendicular to the base

Part A

Square

Part B

Triangle

B. Pyramids

2.56: Right Pyramid

The apex of a right pyramid is exactly above the center of the base.

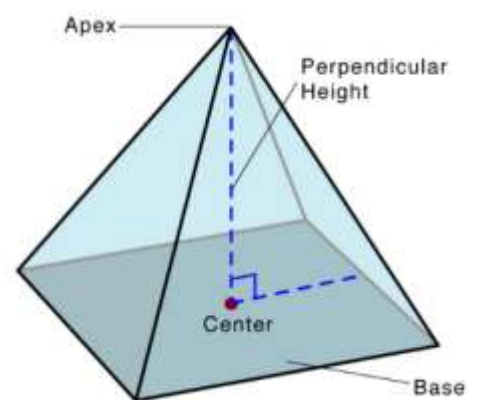
- Height is length of the line segment drawn the apex to the base, perpendicular to the base.
- Slant height is the distance on a lateral face from the center of the face to the apex.

Example 2.57

A right square pyramid has side length of base s and height s .

Find:

- A. the slant height
- B. the angle made by a face with the base
- C. the angle made an edge length (not on the base) of the face with the base



Part A

Take a cross section perpendicular to the base and parallel to a side, passing through the center of the square base: (left diagram).

$$\text{Slant Height} = \sqrt{s^2 + \left(\frac{s}{2}\right)^2} = \sqrt{\frac{5s^2}{4}} = \frac{s\sqrt{5}}{2}$$

Part B

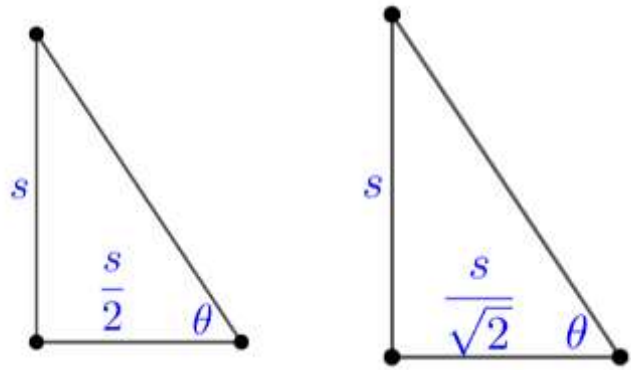
Use the same cross section as in part A.

$$\tan \theta = \frac{\frac{s}{2}}{\frac{s}{2}} = 1 \Rightarrow \theta = \tan^{-1} 1$$

Part C

Take a cross section perpendicular to the square base, and parallel to the diagonal of the base, passing through the center of the square base.

$$\tan \theta = \frac{\frac{s}{2}}{\frac{s}{2\sqrt{2}}} = \sqrt{2} \Rightarrow \theta = \tan^{-1} \sqrt{2}$$



Example 2.58: Pyramid

A square pyramid has all sides length s . What is the angle formed by the slant edge of the pyramid with a base diagonal?

Drop a perpendicular from the vertex of the pyramid to the base. It will touch the base at the center of the square, which is also the point where the diagonals bisect each other.

$$l(\text{Diagonal}) = s\sqrt{2} \Rightarrow \frac{l(\text{Diagonal})}{2} = \frac{s\sqrt{2}}{2}$$

$$\cos \theta = \frac{\text{adj}}{\text{opp}} = \frac{\frac{s\sqrt{2}}{2}}{s} = \frac{s\sqrt{2}}{2} \times \frac{1}{s} = \frac{\sqrt{2}}{2} = \frac{1}{\sqrt{2}} \Rightarrow \cos^{-1} \theta = 45^\circ = \frac{\pi}{4}$$

C. Cubes

2.59: Parallel Lines, Intersecting lines and skew lines (3D Geometry)

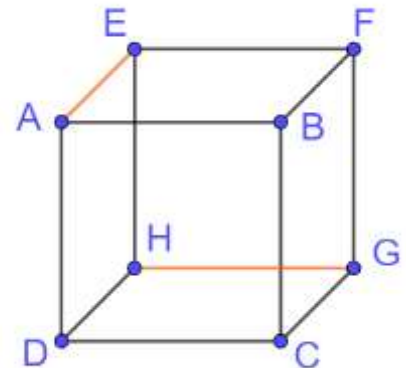
Two lines which go in the same direction are parallel lines.

Two lines which have exactly one point in common are intersecting lines.

Two lines which are neither parallel nor intersecting are skew lines.

For example, in the diagram of a cube:

- AE and HG are skew lines
- AD and DC are intersecting line
- AD and BC are parallel lines



2.60: Intersecting Lines

Two intersecting lines lie in a single plane and the angle between the two lines can be determined by using 2D geometry with respect to that plane.

A plane is determined by a line and a point not on it.

So, two points decide a line, and the direction of the line. A second line which intersects the first has one point in common with the first line.

A point on the second line which is not on the first line determines a plane (in which both the lines lie).

2.61: Types of Diagonals

In a three-dimensional object, a diagonal which

- is contained entirely in a face is a face diagonal.
- Travels through space outside of a face is a space diagonal

Example 2.62

Find the angles between the space diagonals of a cube.

Draw a cube. Without loss of generality, let the side length of the cube be 1

Consider the plane formed by points

A, E, G

And note that C also lies in the same plane.

Note that $AEGC$ forms a rectangle.

$$AE = 1, \quad AC = \sqrt{2}, \quad AG = \sqrt{3}$$

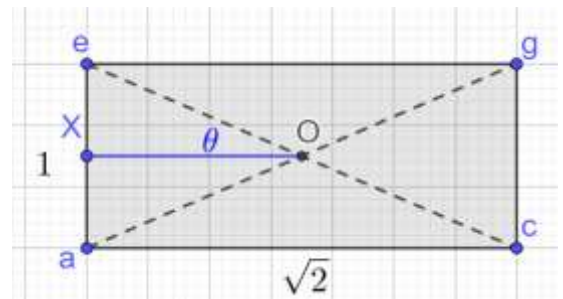
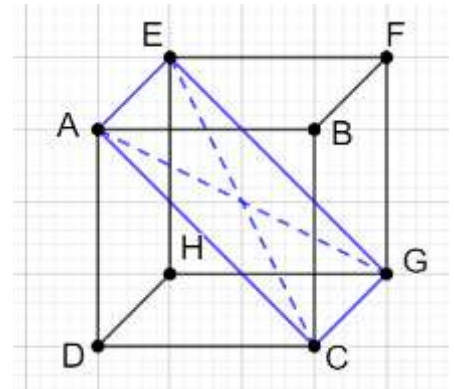
$$XO = \frac{ac}{2} = \frac{\sqrt{2}}{2}$$

$$eX = \frac{ea}{2} = \frac{1}{2}$$

$$\tan \angle eOX = \tan \theta = \frac{\frac{1}{2}}{\frac{\sqrt{2}}{2}} = \frac{1}{\sqrt{2}}$$

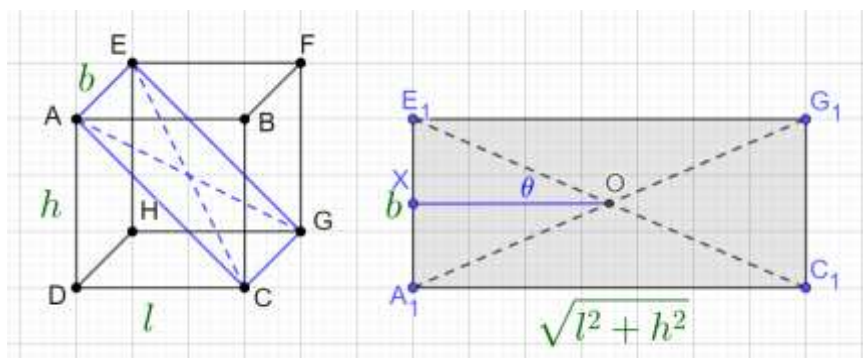
$$\theta = 2 \cdot \tan^{-1} \frac{1}{\sqrt{2}}$$

$$\angle eOa = 2\theta = 2 \cdot \tan^{-1} \frac{1}{\sqrt{2}}$$



Example 2.63

A cuboid has length l , width w , and height h . Find the angles between its space diagonals.



There are two angles that we need to find. By similarity of triangles:

$$\angle E_1 O X = \angle E_1 C_1 A_1$$

$$\tan \theta = \frac{b}{\sqrt{l^2 + h^2}}$$

$$2\theta = 2 \cdot \tan^{-1} \left(\frac{b}{\sqrt{l^2 + h^2}} \right)$$

If 2θ is the smaller angle then, the correct answer will be:

$$2\theta = 2 \cdot \tan^{-1} \left(\frac{\sqrt{l^2 + h^2}}{b} \right)$$

The second angle (without diagram):

Consider rectangle $ABGH$

$$AB = l, AH = \sqrt{h^2 + b^2}$$

$$2\theta = 2 \cdot \tan^{-1} \left(\frac{l}{\sqrt{h^2 + b^2}} \right)$$

Example 2.64: Cuboid

The longest diagonal, the base diagonal and an edge are three sides of a triangle in a cuboid with length l , width w , and height h . Find the angles of that triangle.

$$\text{Cuboid: } \underbrace{\text{Base Diagonal} = \sqrt{l^2 + w^2}}_{\text{Pythagoras}} \Rightarrow \underbrace{\text{Longest Diagonal} = \sqrt{l^2 + w^2 + h^2}}_{\text{Pythagoras}}$$

Example 2.65: Cube

Find the angles of the triangle formed by the longest diagonal, the base diagonal and an edge of a cube.

$$\text{Cube: } \underbrace{\text{Base Diagonal} = \sqrt{2a^2} = a\sqrt{2}}_{\text{Pythagoras}} \Rightarrow \underbrace{\text{Longest Diagonal} = \sqrt{3a^2} = a\sqrt{3}}_{\text{Pythagoras}}$$

D. Tower at center of Circle

This topic is the beginning of questions that require 3D analysis. Till now we were working using one dimension on the ground, and one dimension as height. This is equivalent to taking a cross section. Questions in 3D require multiple cross sections.

Example 2.66: Points on a circle

r cables (where r represents the number of cables) keep the central pole of a tent in position. The ground distance between the anchor points and the foot of the pole is d meters. The angle of elevation from an anchor point to the pole is α . If cable costs $\$D$ per meter, find the cost of the cable in terms of r, d, α and D .

Let

OP be the pole, with O the base and P the top

Let l be the length of a single cable. Then in $\triangle POC$:

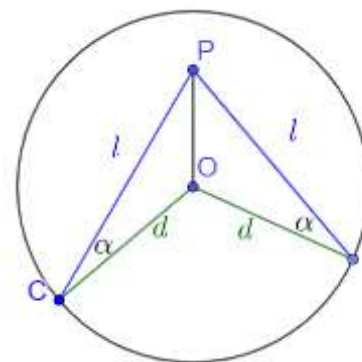
$$\cos \alpha = \frac{d}{l} \Rightarrow l = \frac{d}{\cos \alpha} m$$

Since each of the cables has the same length, the length of r cables is:

$$rl = \frac{rd}{\cos \alpha} m$$

Cost of the cables

$$= \frac{rd}{\cos \alpha} m \times \frac{D}{m} \$ = \frac{rdD}{\cos \alpha} = rdD \sec \alpha \$$$



Example 2.67: Angle subtended by chord

A tower stands at the center of a circular park. A and B are two points on the boundary of the park such that $AB (= a)$ subtends an angle of 60° at the foot of the tower, and the angle of elevation of the top of the tower from A or B is 30° . The height of the tower (in terms of a) is: (JEE-M 2007)

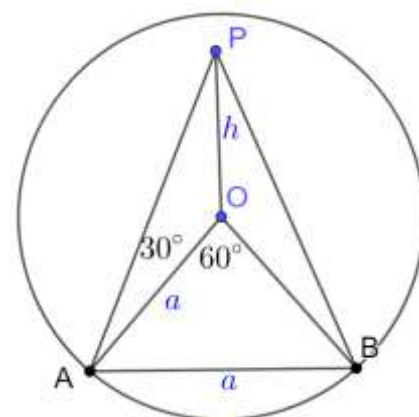
$$\text{Radius} = OA = OB$$

Hence $\triangle AOB$ is isosceles.

$\angle AOB$ is 60° , $\triangle AOB$ is equilateral.

In $\triangle AOP$:

$$\begin{aligned} \tan 30^\circ &= \frac{h}{a} \\ h &= a \tan 30^\circ = \frac{a}{\sqrt{3}} \end{aligned}$$



E. Single Tower (Different angles of elevation)

Example 2.68

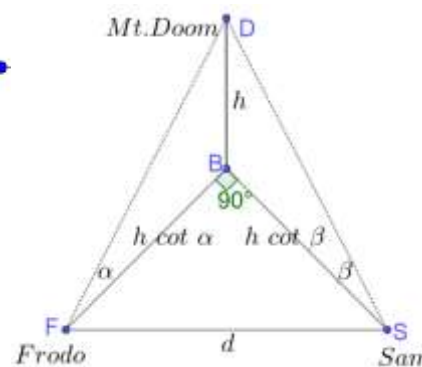
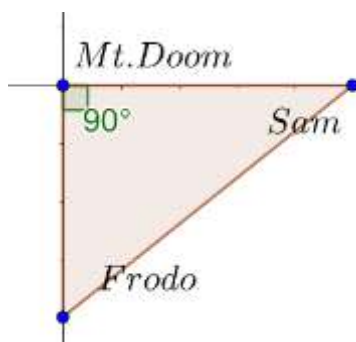
Frodo is staring due North at Mt. Doom, which has height h . Sam is staring due West at Mt. Doom. Frodo and Sam are d units apart, both have the same height above sea level and their angles of elevation to Mt. Doom are α and β respectively. Show that the height of Mount Doom is $\frac{d}{\sqrt{\cot^2 \alpha + \cot^2 \beta}}$.

Mt. Doom is due North of Frodo and due West of Sam: we get a right triangle (left diagram – 2D).

Draw a 3D diagram (right diagram)

$$\text{In } \triangle DBF: \cot \alpha = \frac{BF}{h} \Rightarrow BF = h \cot \alpha$$

$$\text{In } \triangle DBS: \cot \beta = \frac{BS}{h} \Rightarrow BS = h \cot \beta$$



By the Pythagorean Theorem in $\triangle FBS$

$$\underbrace{h^2 \cot^2 \alpha}_{BF^2} + \underbrace{h^2 \cot^2 \beta}_{BS^2} = d^2$$

$$h = \frac{d}{\sqrt{\cot^2 \alpha + \cot^2 \beta}}$$

Example 2.69

The angle of elevation of the top P of a tower from the feet of one person standing due south of the tower is 45° and from the feet of another person standing due west of the tower is 30° . If the height of the tower is 5m , then the distance in meters between the two persons is equal to: (JEE-M 2023)

Draw a diagram. Let:

Foot of the tower = O

Top of the tower = P

Person standing due south be at point B

Person standing due west be at point A

In $\triangle PBO$:

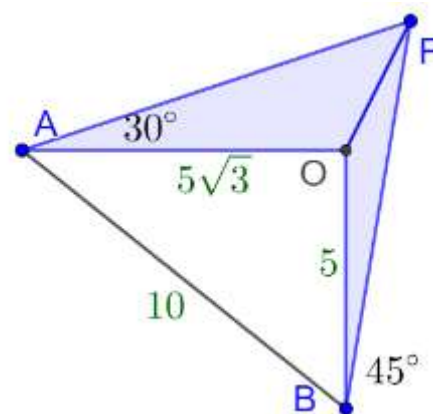
$$\tan \angle PBO = \frac{5}{OB} \Rightarrow OB = \frac{5}{\tan 45^\circ} = \frac{5}{1} = 5$$

In $\triangle PAO$:

$$\tan \angle PAO = \frac{5}{OA} \Rightarrow OA = \frac{5}{\tan 30^\circ} = \frac{5}{\frac{1}{\sqrt{3}}} = 5\sqrt{3}$$

In $\triangle AOB$, by Pythagoras Theorem:

$$AB = \sqrt{OA^2 + OB^2} = \sqrt{5^2 + (5\sqrt{3})^2} = \sqrt{25 + 75} = \sqrt{100} = 10$$

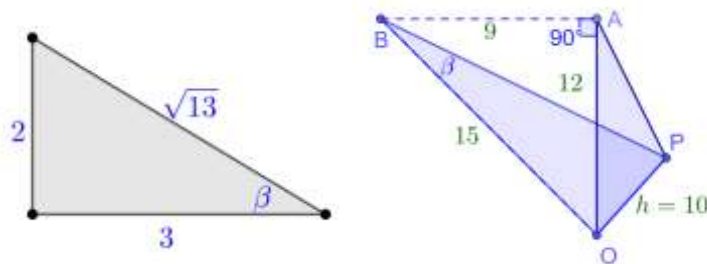


Example 2.70

38: The angle of elevation of the top of a tower from a point A due north of it is α and from a point B at a distance of 9 units due west of A is $\cos^{-1}\left(\frac{3}{\sqrt{13}}\right)$. If the distance of the point B from the tower is 15 units, then $\cot \alpha$ is equal to: (JEE-M 2022)

In $\triangle BAO$, by Pythagorean Triplet:

$$3(3,4,5) = (9,12,15) \Rightarrow OA = 12$$



Take the cos of both sides of $\beta = \cos^{-1}\left(\frac{3}{\sqrt{13}}\right)$ and find $\tan \beta$ using a reference triangle:

$$\cos \beta = \frac{3}{\sqrt{13}} \Rightarrow \tan \beta = \frac{2}{3}$$

$$\text{In } \triangle PBO: \tan \beta = \frac{2}{3} = \frac{h}{15} \Rightarrow h = 10$$

$$\text{In } \triangle POA: \cot \angle PAO = \cot \alpha = \frac{12}{10} = \frac{6}{5}$$

*Example 2.71

A horizontal park is in the shape of a triangle OAB with $AB = 16$. A vertical lamp post OP is erected at the point O such that $\angle PAO = \angle PBO = 15^\circ$, and $\angle PCO = 45^\circ$, where C is the midpoint of AB . Then OP^2 is equal to: (JEE-M 2022)

In $\triangle PAO$ and $\triangle PBO$:

$$\angle POA = \angle POB = 90^\circ$$

$$\angle PAO = \angle PBO = 15^\circ$$

$$PO = PO$$

$$\triangle PAO \cong \triangle PBO \text{ (SAA)}$$

$$OA = OB \text{ (CPCT)}$$

Hence:

$$\triangle OAB \text{ is isosceles} \Rightarrow \angle OCA = 90^\circ$$

In right $\triangle AOC$, by the Pythagorean Theorem:

$$\underline{OA^2 = OC^2 + AC^2} \quad \text{Equation I}$$

Note that:

$$\text{In } \triangle POC: \angle PCO = \angle OPC = 45^\circ \Rightarrow \underline{OP = OC} \quad \text{Equation II}$$

$$\text{In } \triangle PAO: \tan 15^\circ = \frac{OP}{OA} \Rightarrow OA = \frac{OP}{\tan 15^\circ} = \underline{OP \cot 15^\circ} \quad \text{Equation III}$$

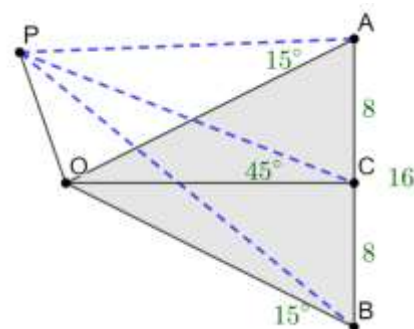
$$AC = 8 \Rightarrow \underline{AC^2 = 64} \quad \text{Equation IV}$$

Substitute Equations II, III and IV in Equation I:

$$(OP \cot 15^\circ)^2 = OP^2 + 64 \Rightarrow OP^2 = \frac{64}{\cot^2 15^\circ - 1}$$

Substitute $\cot^2 15^\circ - 1 = (2 + \sqrt{3})^2 - 1 = 4 + 4\sqrt{3} + 3 - 1 = 6 + 4\sqrt{3}$:

$$OP^2 = \frac{64}{\cot^2 15^\circ - 1} = \frac{64}{6 + 4\sqrt{3}} = \frac{32}{3 + 2\sqrt{3}} = \frac{32(3 - 2\sqrt{3})}{9 - 12} = \frac{32(2\sqrt{3} - 3)}{3}$$



Pending

Example 2.72

54: ABC is a triangular park with $AB = AC = 100$ meters. A vertical tower is situated at the midpoint of BC. If the angles of elevation of the top of the tower at A and B are $\cot^{-1} 3\sqrt{2}$ and $\csc^{-1} 2\sqrt{2}$ respectively, then the height of the tower (in meters) is: (JEE-M 2018, 2019)

F. Multiple Towers

Example 2.73

A surveyor wanting to find the width of a river plants two poles (of equal height) d meters apart on one bank of the river. The surveyor's assistant standing directly opposite one of the poles, measures the angles of elevation to the poles to be α° and β° , with $\alpha > \beta$. Determine the width w of the river.

Combined Diagram

Let the poles be:

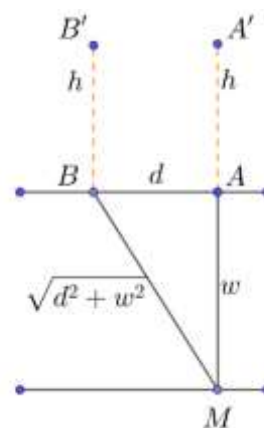
AA' : With A the base and A' the top

BB' : With B the base and B' the top

Draw a 3D diagram. Let the height of each pole be h . the assistant be standing at point M.

In $\triangle MAB$, by the Pythagorean Theorem:

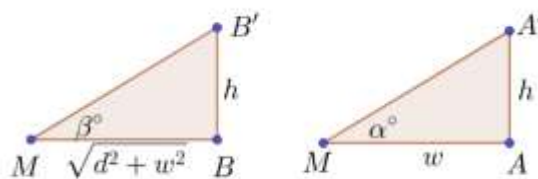
$$MB = \sqrt{d^2 + w^2}$$



Split the Triangles

$$\text{From } \triangle MAA': \tan \alpha = \frac{h}{w} \Rightarrow \underbrace{h = w \tan \alpha}_{\text{Equation I}}$$

$$\text{From } \triangle MBB': \tan \beta = \frac{h}{\sqrt{d^2 + w^2}} \Rightarrow \underbrace{h = \sqrt{d^2 + w^2} \tan \beta}_{\text{Equation II}}$$



From Equation I and II:

$$w \tan \alpha = \sqrt{d^2 + w^2} \tan \beta$$

Square both sides:

$$\begin{aligned} w^2 \tan^2 \alpha &= (d^2 + w^2) \tan^2 \beta \\ w^2 (\tan^2 \alpha - \tan^2 \beta) &= d^2 \tan^2 \beta \\ w &= \frac{\sqrt{\tan^2 \beta d^2}}{\sqrt{\tan^2 \alpha - \tan^2 \beta}} = \frac{d \tan \beta}{\sqrt{\tan^2 \alpha - \tan^2 \beta}} \end{aligned}$$

G. Cones

2.74: Making a Cone from a Sector

Example 2.75

A three-quarter sector of a circle of radius 4 inches together with its interior can be rolled up to form the lateral

surface area of a right circular cone by taping together along the two radii shown. What is the volume of the cone in cubic inches? (AMC 10B 2020/10, AMC 12B 2020/9)

Example 2.76

Surface Area of a Cone

2.4 Bearings and Coordinate Geometry

A. Bearings

2.77: True Bearings

- True bearings give the direction from north.
- True bearings are always written using three digits.

2.78: Reverse Bearings

If the bearing from A to B is α , then the bearing from B to A is:

- $\alpha + 180$ if $0^\circ \leq \alpha < 180^\circ$
- $\alpha - 180$ if $180^\circ \leq \alpha < 360^\circ$

- Getting a reverse bearing means going the opposite direction.
- Hence, it is equivalent to a 180° turn.

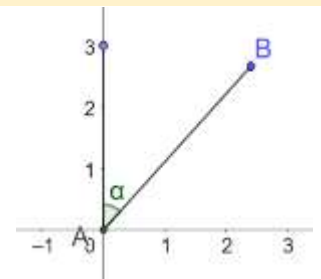
Example 2.79

In each part below, draw a sketch, and find the reverse bearing. If more than one bearing is given, find the reverse bearing for each.

- A. The bearing from A to B is 042° . Sketch a diagram.
- B. The bearing from A to B is 123° . The bearing from B to C is 22° . The bearing from C to A is 260° .

Example 2.80

- A. A whaler leaves port P at a bearing of 108° , and travels for 50 miles to point W, where it harpoons a whale. It then changes direction, and travels 120 miles at a bearing of 198° to reach A. Find the bearing to home port for the whaler.
- B. (Bearings Type I)
- C. (Bearings Type I)



Draw a diagram.

By co-interior angles, the angle at point A and the angle at point P are supplementary:

$$\text{The angle at A} = 180 - 108 = 72^\circ$$

By angles around a point:

$$\angle PWA = 360 - 72 - 198 = 90^\circ$$

(5,12,13) is a Pythagorean triplet:

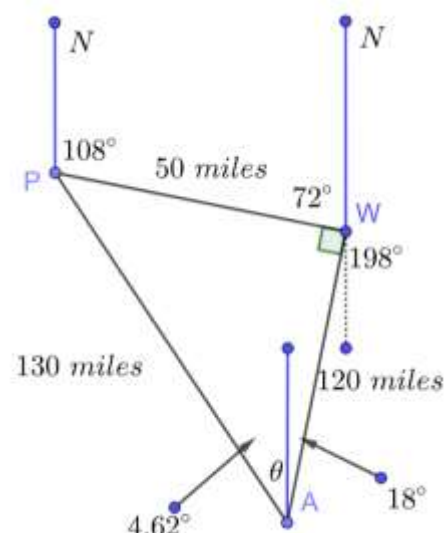
$$10(5,12,13) = (50,120,130)$$

$$\sin \theta = \frac{50}{130} = \frac{5}{13} \Rightarrow \theta = \sin^{-1} \frac{5}{13} \approx 22.62^\circ$$

$$22.62 - 18 = 4.62$$

And finally, the bearing from A to P is:

$$360 - 4.62 = 355.38^\circ$$



B. Coordinate Geometry

Example 2.81

The adjoining diagram shows a circle on the coordinate plane with its center three units to the right and five units above the origin. Find the coordinates of points T, Q and P.

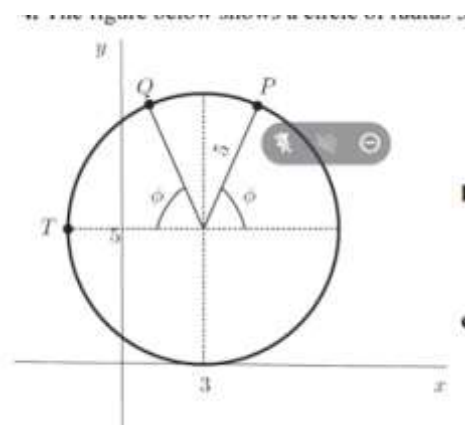
The coordinates of the center of the circle are (3,5)

$$\sin \phi = \frac{Y}{5} \Rightarrow Y = 5 \sin \phi \Rightarrow P_y = 5 \sin \phi + 5$$

$$\cos \phi = \frac{X}{5} \Rightarrow X = 5 \cos \phi \Rightarrow P_x = 5 \cos \phi + 3$$

$$P(x, y) = (5 \cos \phi + 3, 5 \sin \phi + 5)$$

$$Q(x, y) = (3 - 5 \cos \phi, 5 \sin \phi + 5)$$



C. Applications

Example 2.82: Applications

A. (Challenge) Distance between Parallel Lines

83 Examples